INFORMATICS INSTITUTE OF TECHNOLOGY

UNIVERSITY OF WESTMINSTER

**Informatics Institute of Technology**

Department of Computing

(B.Eng.) in Software Engineering

**Module: 5COSC007C - Object Oriented Programming**

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# COURSEWORK 1


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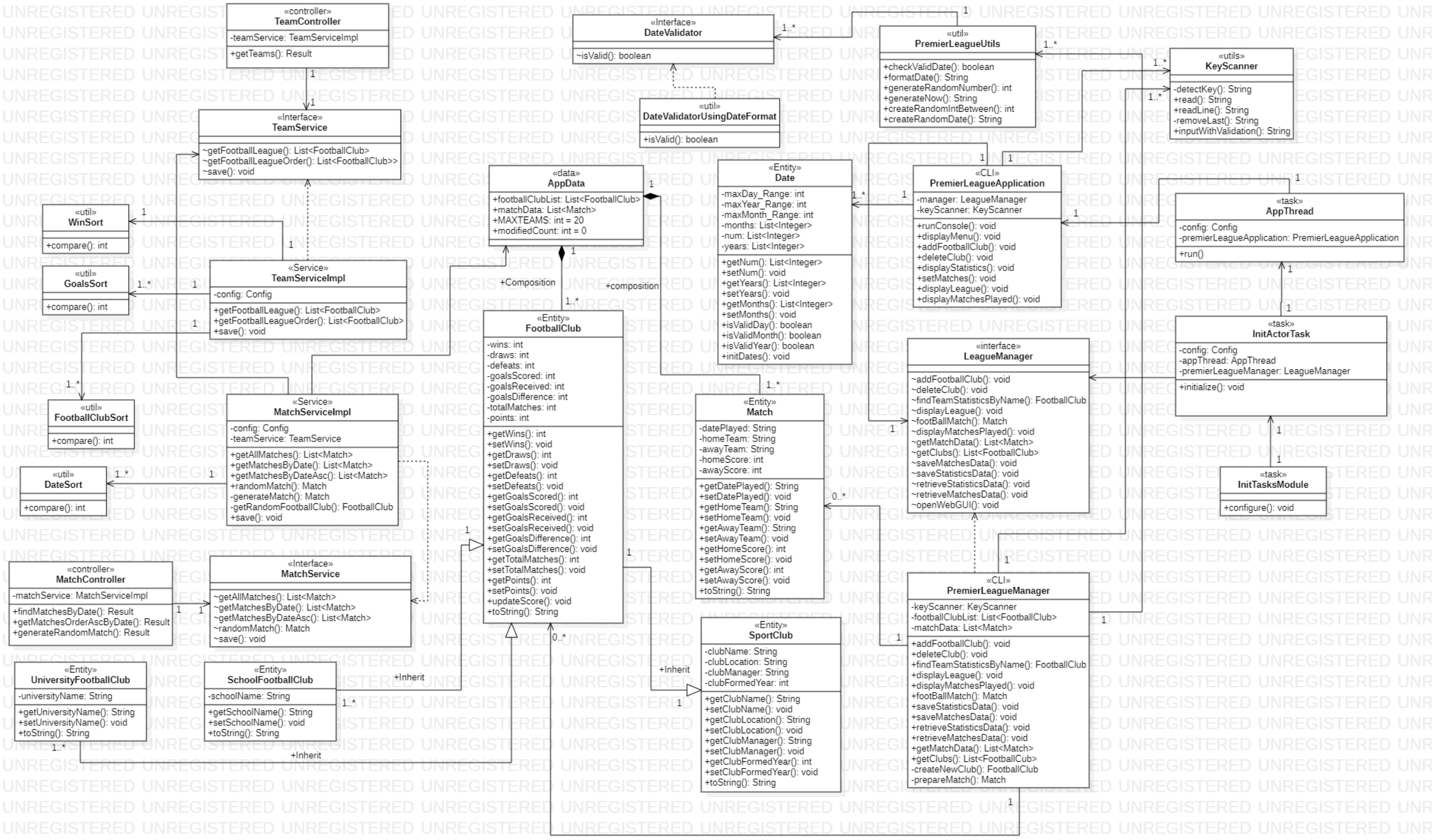
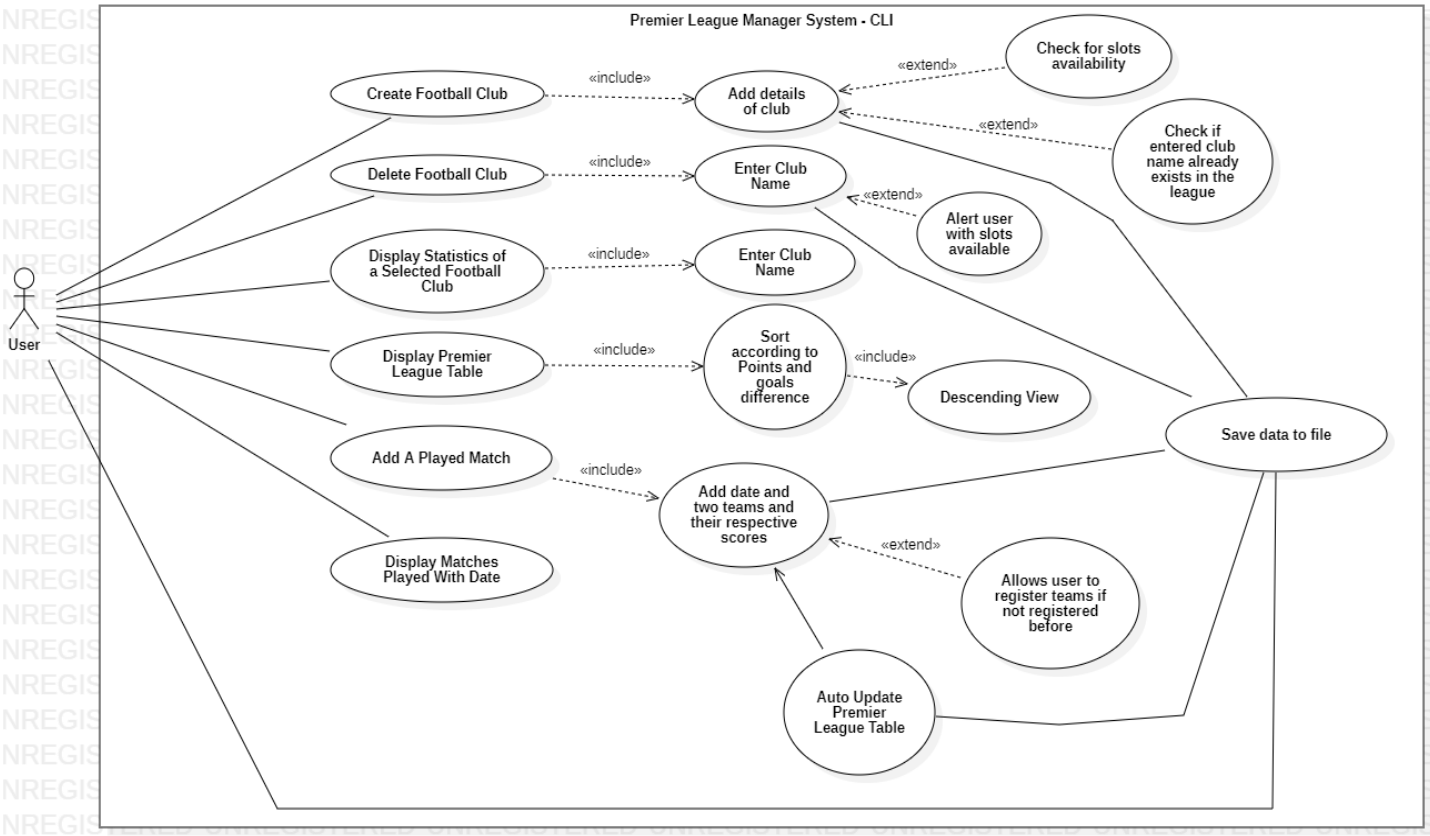
# 1) Class Diagrams and Use Case Diagrams

## 1.1) Class Diagrams



## 1.2) Use Case Diagrams

### 1.2.1) Use case diagram for CLI

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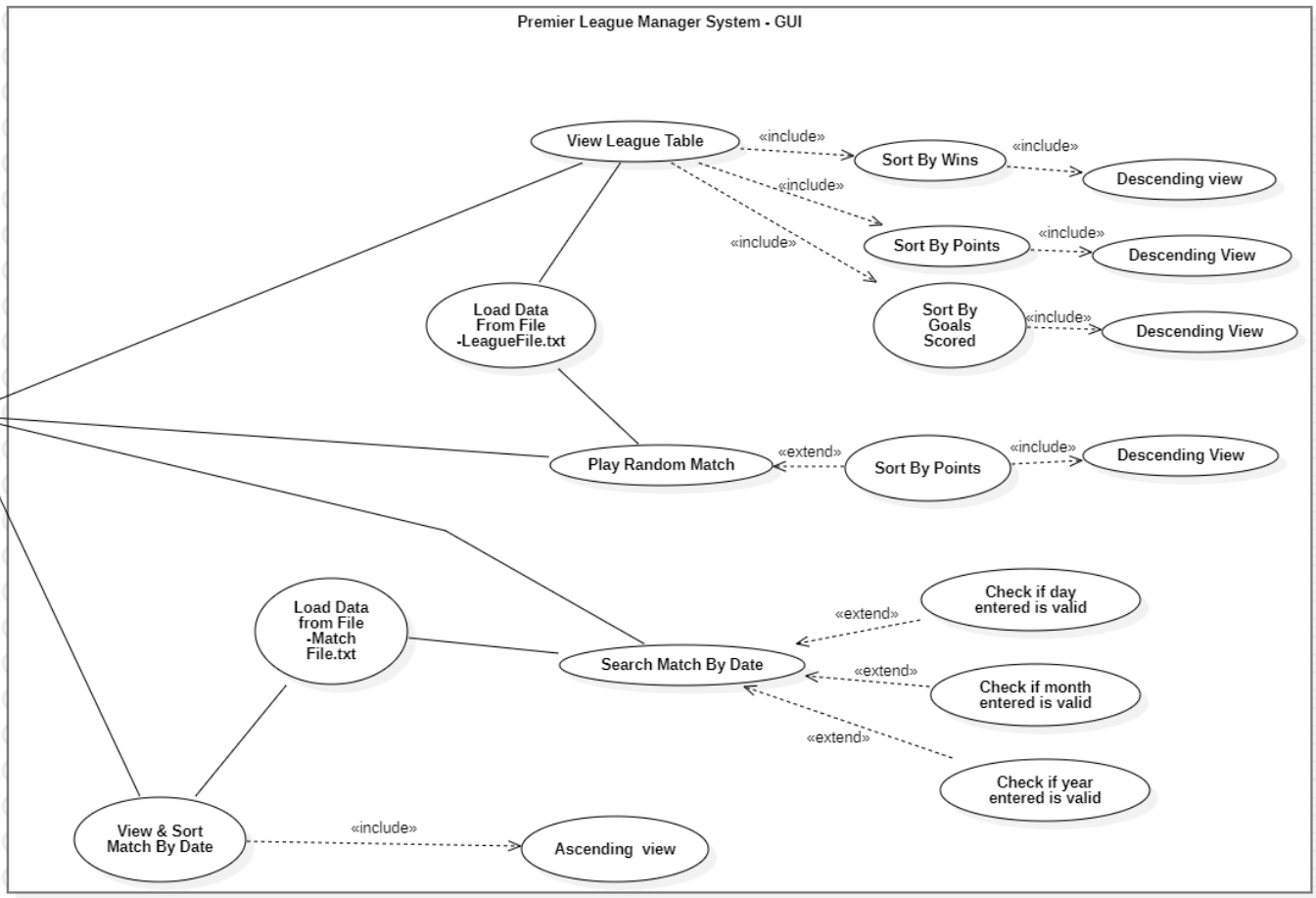
### 1.2.2) Use case diagram for GUI

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# 2) Play Framework - Java

### 2.1.1) SportsClub Class

```
package entities.team;

import java.io.Serializable;

/**
* Class that handle SportClub data
*/
public class SportsClub implements Serializable {
   private String clubName;
   private String clubLocation;
   private String clubManager;
   private int clubFormedYear;

   public SportsClub() {

   }

   public SportsClub(String clubName, String clubLocation, String clubManager, int
clubFormedYear) {
      this.clubName = clubName;
      this.clubLocation = clubLocation;
      this.clubManager = clubManager;
      this.clubFormedYear = clubFormedYear;

   }


   public String getClubName() {
      return clubName;
   }

   public void setClubName(String clubName) {
      this.clubName = clubName;
   }

   public String getClubLocation() {
      return clubLocation;
   }

   public void setClubLocation(String location) {
      this.clubLocation = location;
   }
```

```
    public String getClubManager() {
        return clubManager;
    }

    public void setClubManager(String clubManager) {
        this.clubManager = clubManager;
    }

    public int getClubFormedYear() {
        return clubFormedYear;
    }

    public void setClubFormedYear(int clubFormedYear) {
        this.clubFormedYear = clubFormedYear;
    }


    @Override
    public String toString() {
        return "SportsClub{clubName=" + clubName + ", clubLocation=" + clubLocation +
", clubManager='" + clubManager +
                ", clubFormedYear=" + clubFormedYear + "}";

    }
}
```

### 2.1.2) FootballClub Class

```
package entities.team;

/**
* Class that handle Football Club data
*/
public class FootballClub extends SportsClub {
    private int wins;
    private int draws;
    private int defeats;
    private int goalsScored;
    private int goalsReceived;
    private int goalsDifference;
    private int totalMatches;
    private int points;
```



```
    public FootballClub(){
```

```
    }
    public FootballClub(String name, String location,String clubManager,int
clubFormedYear) {
        super(name, location,clubManager,clubFormedYear);

    }
    public FootballClub(String name,String location,String clubManager,int
clubFormedYear,int wins,int defeats,int draws,int goalsScored,int goalsReceived,int
totalMatches,int points){
        super(name, location,clubManager,clubFormedYear);
        this.wins = wins;
        this.draws = draws;
        this.defeats = defeats;
        this.goalsScored = goalsScored;
        this.goalsReceived = goalsReceived;
        this.totalMatches = totalMatches;
        this.points = points;

    }


    public int getWins() {
        return wins;
    }

    public void setWins(int wins) {
        this.wins+= wins;
        this.points+=(wins*3);
        this.totalMatches+=wins;
    }

    public int getDraws() {
        return draws;
    }

    public void setDraws(int draws) {
        this.totalMatches+=draws;
        this.draws+= draws;
        this.points+=draws;

    }

    public int getDefeats() {
        return defeats;
    }

    public void setDefeats(int defeats) {
```

```
    this.totalMatches+=defeats;
    this.defeats =this.defeats+ defeats;

}


public int getGoalsScored() {
    return goalsScored;
}

public void setGoalsScored(int goalsScored) {
    this.goalsScored= this.goalsScored+goalsScored;
    this.goalsDifference=goalsScored-goalsReceived;


}

public int getGoalsReceived() {
    return goalsReceived;
}

public void setGoalsReceived(int goalsReceived) {

    this.goalsReceived= this.goalsReceived+goalsReceived;
    this.goalsDifference=goalsScored-goalsReceived;

}

public int getGoalsDifference() {
    return goalsDifference;
}

public void setGoalsDifference(int goalsDifference) {
    this.goalsDifference=this.goalsScored - this.goalsReceived;
}

public int getTotalMatches() {
    return totalMatches;
}

public void setTotalMatches(int totalMatches) {
    this.totalMatches = totalMatches;
}

public int getPoints() {
    return points;
}
```

```
    public void setPoints(int points) {
        this.points+= points;
    }


    public void updateScore(int goalscore, int goalreceive) {
        this.setGoalsScored(goalscore);
        this.setGoalsReceived(goalreceive);
        this.setGoalsDifference(goalscore-goalreceive);
        if (goalscore == goalreceive) {
            setDraws(1);
        }
    }

    @Override
    public String toString() {
        return  super.toString()+" FootballClub{wins=" + wins + ", draws=" + draws + ",
defeats=" + defeats + ", goalsScored=" + goalsScored +
                ", goalsReceived=" + goalsReceived + ", goalsDifference=
"+goalsDifference+", totalMatches=" + totalMatches + ", points=" + points + "}";
    }



}
```

### 2.1.3) UniversityFootballClub class

```
package entities.team;

/**
 * Class that handle UniversityFootball Club data
 */
public class UniversityFootballClub extends FootballClub{
    private String universityName;

    public UniversityFootballClub(){

    }
    public UniversityFootballClub(String universityName) {
        this.universityName = universityName;
    }

    public String getUniversityName() {
        return universityName;
```

```
    }

    public void setUniversityName(String universityName) {
        this.universityName = universityName;
    }

    @Override
    public String toString() {
        return  super.toString()+" UniversityFootballClub{universityName= " +
universityName +"}";
    }
}
```

### 2.1.4) SchoolFootballClub class

```
package entities.team;

/**
 * Class that handle SchoolFootball Club data
 */
public class SchoolFootballClub extends FootballClub{
    private String schoolName;

    public SchoolFootballClub(){

    }

    public SchoolFootballClub(String schoolName) {
        this.schoolName = schoolName;
    }

    public String getSchoolName() {
        return schoolName;
    }

    public void setSchoolName(String schoolName) {
        this.schoolName = schoolName;
    }

    @Override
    public String toString() {
        return  super.toString()+" SchoolFootballClub{schoolName= " + schoolName +"}";
    }
}
```

### 2.1.5) Date Class

```
package entities.match;

import java.io.Serializable;
import java.util.ArrayList;
import java.util.List;

/**
* Class that handle Date data
*/
public class Date implements Serializable {

   private static final int maxDay_Range = 30;
   private static final int maxYear_Range = 2020;
   private static final int maxMonth_Range = 12;

   private List<Integer> months = new ArrayList<>();
   private List<Integer> num = new ArrayList<>();
   private List<Integer> years = new ArrayList<>();

   public Date() {
      initDates();
   }

   public List<Integer> getNum() {
      return num;
   }

   public void setNum(List<Integer> num) {
      this.num = num;
   }

   public List<Integer> getYears() {
      return years;
   }

   public void setYears(List<Integer> years) {
      this.years = years;
   }

   public List<Integer> getMonths() {
      return months;
   }

   public void setMonths(List<Integer> months) {
      this.months = months;
   }
```

```
    public boolean isValidDay(int day) {
        int key = 0;

        if (num.contains(day)) {
            if (day <= maxDay_Range && day > 0) {
                key++;
            }
        }
        return key > 0;
    }

    public boolean isValidMonth(int str) {
        int key = 0;

        if (str <= maxMonth_Range && str > 0) {
            key++;
        }

        return key > 0;
    }

    public boolean isValidYear(int str) {
        int key = 0;
        if (years.contains(str)) {
            if (str <= maxYear_Range) {
                key++;
            }
        }
        return key > 0;
    }

    private void initDates() {
        for (int i = 0; i < 31; i++) {
            num.add(i);
        }
        for (int i = 1986; i < 2022; i++) {
            years.add(i);
        }
    }
}
```

### 2.1.6) Match Class

```
package entities.match;

import java.io.Serializable;

/**
* Class that handle Match data
*/
public class Match implements Serializable {
   private String datePlayed;
   private String homeTeam;
   private String awayTeam;
   private int homeScore;
   private int awayScore;

   public Match() {
   }

   public Match(String datePlayed, String homeTeam, String awayTeam, int
homeScore, int awayScore) {
      this.datePlayed = datePlayed;
      this.homeTeam = homeTeam;
      this.awayTeam = awayTeam;
      this.homeScore = homeScore;
      this.awayScore = awayScore;
   }


   public String getDatePlayed() {
      return datePlayed;
   }

   public void setDatePlayed(String datePlayed) {
      this.datePlayed = datePlayed;
   }

   public String getHomeTeam() {
      return homeTeam;
   }

   public void setHomeTeam(String homeTeam) {
      this.homeTeam = homeTeam;
   }

   public String getAwayTeam() {
      return awayTeam;
   }
```

```
    public void setAwayTeam(String awayTeam) {
        this.awayTeam = awayTeam;
    }

    public int getHomeScore() {
        return homeScore;
    }

    public void setHomeScore(int homeScore) {
        this.homeScore = homeScore;
    }

    public int getAwayScore() {
        return awayScore;
    }

    public void setAwayScore(int awayScore) {
        this.awayScore = awayScore;
    }


    @Override
    public String toString() {
        return "Match{datePlayed=" + datePlayed + ", homeTeam=" + homeTeam + ",
awayTeam=" + awayTeam + ", homeScore=" + homeScore +", awayScore=" +
awayScore + "}";
    }


}
```

## 2.2) utils

### 2.2.1) DateSort class

```
package utils.sort;

import entities.match.Match;

import java.text.DateFormat;
import java.text.ParseException;
import java.text.SimpleDateFormat;
import java.util.Comparator;
```

```
/**
* A Class to sort match list by date
*/
public class DateSort implements Comparator<Match> {

        /**
        * compare two match data by date
        * @param o1 Match
        * @param o2 Match
        * @return int
        */
   @Override
   public int compare(Match o1, Match o2) {
      DateFormat f = new SimpleDateFormat("dd/MM/yyyy");
      try {
         return f.parse(o1.getDatePlayed()).compareTo(f.parse(o2.getDatePlayed()));
      } catch (ParseException e) {
         throw new IllegalArgumentException(e);
      }
   }
}
```

### 2.2.2)FootballClubSort class

```
package utils.sort;

import entities.team.FootballClub;
import java.util.Comparator;

/**
* A Class to sort football club list by points
*/
public class FootballClubSort implements Comparator<FootballClub> {

        /**
        * compare 2 football club data by Points
        * @param o1 Football club
        * @param o2 Football club
        * @return int
        */
   @Override
   public int compare(FootballClub o1, FootballClub o2) {
     int rtrn;
      if(o1.getPoints() == o2.getPoints()){
```

```
            if(o1.getGoalsDifference() == o2.getGoalsDifference()){
                rtrn = 0;
            }else if(o1.getGoalsDifference() < o2.getGoalsDifference()){
                rtrn = 1;
            }else{
                rtrn = -1;
            }
        }
        else if(o1.getPoints() < o2.getPoints()){
            rtrn = 1;
        }
        else if(o1.getPoints() > o2.getPoints()){
            rtrn = -1;
        }
        else{
            rtrn = 0;
        }
        return rtrn;
    }
}
```

### 2.2.3) GoalsSort class

```
package utils.sort;

import entities.team.FootballClub;
import java.util.Comparator;

/**
* A Class to sort football club list by goals score
*/
public class GoalsSort implements Comparator<FootballClub> {

            /**
            * compare 2 football club data by goals score
            * @param ft Football club
            * @param ft1 Football club
            * @return int
            */
    @Override
    public int compare(FootballClub ft, FootballClub ft1) {
        if (ft.getGoalsScored() == ft1.getGoalsScored())
            return 0;
        else if (ft.getGoalsScored() > ft1.getGoalsScored())
            return -1;
        else
```

```
            return 1;
        }
    }
```

### 2.2.4) WinSort class

```
package utils.sort;

import entities.team.FootballClub;
import java.util.Comparator;

/**
* A Class to sort football club list by wins
*/
public class WinSort implements Comparator<FootballClub> {

            /**
             * compare 2 football club data by wins
             * @param ft Football club
             * @param ft1 Football club
             * @return int
             */
    @Override
    public int compare(FootballClub ft, FootballClub ft1) {
        if (ft.getWins() == ft1.getWins())
            return 0;
        else if (ft.getWins() > ft1.getWins())
            return -1;
        else
            return 1;
    }
}
```

### 2.2.5) DateValidator Interface

```
package utils.validator.date;

public interface DateValidator {
    boolean isValid(String dateStr);
}
```

### 2.2.6) DateValidatorUsingDateFormat class

```
package utils.validator.date;

import java.time.LocalDate;
import java.time.format.DateTimeFormatter;
import java.time.format.DateTimeParseException;
import java.time.format.ResolverStyle;
import java.util.Locale;

/**
* A class that handle date validation
*/
public class DateValidatorUsingDateFormat implements DateValidator {
   private String dateFormat;

   public DateValidatorUsingDateFormat(String dateFormat) {
      this.dateFormat = dateFormat;
   }

   /**
   * check whether date valid or not
   * @param dateStr date input
   * @return boolean
   */
   @Override
   public boolean isValid(String dateStr) {
      try {
         if (dateStr.matches("([0-9]{1,2})/([0-9]{1,2})/([0-9]{4})")) {
            LocalDate.parse(dateStr,
                  DateTimeFormatter
                        .ofPattern(this.dateFormat));

         } else {
            return false;
         }
      } catch (DateTimeParseException e) {
         return false;
      }
      return true;
   }
}
```

### 2.2.6) KeyScanner class

```
package utils;

import java.io.IOException;

/**
* handle user input from console
* this class was made because CLI App runs in another Thread of Play App.
* Play Console is used by Play App to log activities of Web App.
* So, user input from another thread cannot be displayed on Play Console due to
blocking
*/
public class KeyScanner{

   /**
   * capture every keypress from user input, store it in variable
   * @param breakline option to make new line or not after enter key is pressed
   * @return String
   */
         private String detectKey(boolean breakline) throws IOException{
      String input = "";
      char c;
      int charCode = 0;

      while(true){
         charCode = System.in.read();

         if(charCode == 13){
            System.out.print("\n");
            break;
         }else if((charCode == 127) | (charCode == 8)){
            System.out.print("\b");
            input = removeLast(input);
         }else{
            c = (char) charCode;
            System.out.print(c);
            input += c;
         }

      }

      return input;
   }

   /**
   * read input user without newline
   * @return String
```

```
   */
   public String read() throws IOException{
          return detectKey(false);
   }

   /**
   * read input user with newline
   * @return String
   */
   public String readLine() throws IOException{
          return detectKey(true);
   }

   /**
   * remove last character of String
   * @param str word or String input
   * @return String
   */
   private String removeLast(String str) {
      if (str != null && str.length() > 0) {
         str = str.substring(0, str.length() - 1);
      }
      return str;
   }

   /**
   * Input with validation and error message
   * @param command
   * @param errorMessage
   * @return String
   */
   public String inputWithValidation(String command, String errorMessage) throws
IOException{
      String in = "";
      boolean err = false;

      while(in.equalsIgnoreCase("")){
         if (err) {
            System.out.println(errorMessage);
            err = false;
         }

         System.out.print(command);
         in = this.readLine();

         if (in.equalsIgnoreCase("")) {
            err = true;
         }
```

```
        }

        return in;
    }

}
```

### 2.2.7) PremierLeagueUtils class

```
package utils;

import utils.validator.date.DateValidator;
import utils.validator.date.DateValidatorUsingDateFormat;

import java.time.LocalDate;
import java.time.format.DateTimeFormatter;
import java.util.Random;

/**
 * Utility class for Premier League application.
 */

public final class PremierLeagueUtils {

    /**
    * check valid date
    * @param date
    * @return boolean
    */
    public static boolean checkValidDate(String date) {
        DateValidator validator = new DateValidatorUsingDateFormat("d/M/yyyy");
        return validator.isValid(date);
    }

    /**
    * change format of date, if day and month have only one character, it will be filled
with zero
    * @param date String of date
    * @return String
    */
    public static String formatDate(String date) {
        String[] split = date.split("/");
        split[0] = (split[0].length() == 1) ? "0" + split[0] : split[0];
        split[1] = (split[1].length() == 1) ? "0" + split[1] : split[1];
        return split[0] + "/" + split[1] + "/" + split[2];
    }
```

```
    /**
    * generate random number with bound
    * @param bound number of bound
    * @return int
    */
    public static int generateRandomNumber(int bound) {
        Random rand = new Random();
        return rand.nextInt(bound);
    }

    /**
    * generate random year
    * @return String
    */
    public static String generateNow() {
        return createRandomDate(2019, 2020);
    }

    /**
    * genererate random number between 2 numbers, start and end
    * @param start
    * @param end
    * @return int
    */
    public static int createRandomIntBetween(int start, int end) {
        return start + (int) Math.round(Math.random() * (end - start));
    }

    /**
    * generate random date between 2 years
    * @param startYear
    * @param endYear
    * @return String
    */
    public static String createRandomDate(int startYear, int endYear) {
        int day = createRandomIntBetween(1, 28);
        int month = createRandomIntBetween(1, 12);
        int year = createRandomIntBetween(startYear, endYear);
        return LocalDate.of(year, month, day)
                .format(DateTimeFormatter.ofPattern("dd/MM/yyyy"));
    }

}
```

## 2.3) data

### 2.3.1) AppData class

```
package data;

import entities.match.Match;
import entities.team.FootballClub;

import java.util.ArrayList;
import java.util.List;

/**
* Store Array List data of football clubs registered and data of played matches
*/
public final class AppData {
   public static List<FootballClub> footballClubList = new ArrayList<>(); //arraylist that
stores football clubs data
   public static List<Match> matchData = new ArrayList<>();  //arraylist that stores
matches data
   public static final int MAX_TEAMS = 20; //maximum number of clubs in the league
   public static int modifiedCount = 0; // This variable is used to note whether data has
modified or not by adding football club, set match or deleting club


}
```

## 2.4) cli

### 2.4.1) LeagueManager Interface

```
package cli;

import com.google.inject.ImplementedBy;
import entities.match.Match;
import entities.team.FootballClub;
import java.util.List;

@ImplementedBy(PremierLeagueManager.class)
public interface LeagueManager {
   void addFootballClub(FootballClub footballClub);

   void deleteClub(String name);

   FootballClub findTeamStatisticsByName(String name);
```

```
    void displayLeague();

    Match footBallMatch(String homeTeamName, int homeScore, String
awayTeamName, int awayScore, String dateFormat);

    void displayMatchesPlayed();

    List<Match> getMatchData();

    List<FootballClub> getClubs();

    void saveMatchesData(String filename);

    void saveStatisticsData(String filename);

    void retrieveStatisticsData(String fileName);

    void retrieveMatchesData(String fileName);

    void openWebGUI();

}
```

### 2.4.2) PremierLeagueManager class

```
package cli;

import entities.match.Match;
import entities.team.FootballClub;
import utils.PremierLeagueUtils;
import utils.sort.FootballClubSort;
import data.AppData;
import utils.KeyScanner;

import java.io.*;
import java.util.List;


/**
* This Class is responsible to manage logic of CLI application
 and to save data and retrieve data from text file
*/

public class PremierLeagueManager implements LeagueManager {
    private KeyScanner keyScanner = new KeyScanner();
    private List<FootballClub> footballClubList = AppData.footballClubList;
```

```
private List<Match> matchData = AppData.matchData;

/**
* get all match
*/
@Override
public List<Match> getMatchData() {
   return matchData;
}

/**
* get all football club
*/
@Override
public List<FootballClub> getClubs() {
   return footballClubList;
}
```

### *2.4.2.1) Creating football Club*

```
 /**
* add football club to ArrayList
* @param footballClub FootballClub instance
*/
@Override
public void addFootballClub(FootballClub footballClub) {
   if (footballClubList.isEmpty()) {
      this.footballClubList.add(footballClub);
      System.out.println("\"" + footballClub.getClubName() + "\" club is successfully
registered to the league ");
   }
            else if (this.footballClubList.size() >= AppData.MAX_TEAMS){
                  System.out.println("All the slots have been filled");
            }
            else {
      boolean findTeam = false;
      for (FootballClub footballClub1 : getClubs()) {
         if (footballClub1.getClubName().toLowerCase()
               .equals(footballClub.getClubName().toLowerCase())) {
            findTeam = true;
            break;
         }
      }
      if (!findTeam) {
         AppData.modifiedCount += 1;
         this.footballClubList.add(footballClub);
         System.out.println("\"" + footballClub.getClubName() + "\" club is successfully
registered to the league ");
```

```
                                    System.out.println("Free Slots Remaining :- " +
(AppData.MAX_TEAMS - this.footballClubList.size()));
            } else {
                System.out.println("The team is already in the list");
            }
        }
    }
```

### *2.4.2.2) Deleting football Club*

```
    /**
    * remove football club by club name
    * @param name club name
    */
    @Override
public void deleteClub(String name) {
        if (footballClubList.isEmpty()) {
            System.out.println("No teams are available in the football team list!");
        } else {
            boolean found = false;
            for (FootballClub club : footballClubList) {
                String cname = club.getClubName();
                if (cname.equals(name)) {
                    found = true;
                    footballClubList.remove(club);
                    System.out.println("\"" + name + "\" club is successfully deleted from the
list");
                    AppData.modifiedCount += 1;
          System.out.println("Free Slots Remaining :- " + (AppData.MAX_TEAMS -
this.footballClubList.size()));
                    break;
                }
            }
            if (!found) {
                System.out.println("No such club exists in the list");
            }
        }

    }
```

### *2.4.2.3) Displaying Statistics of a selected football Club*

```
    /**
    * find team statistics by name
    * @param name club name
    */
    @Override
```

```
public FootballClub findTeamStatisticsByName(String name) {
    if (footballClubList.isEmpty()) {
        System.out.println("No teams are present in the list.");
    } else {
        try {
            for (FootballClub ftc : footballClubList) {
                String cname = ftc.getClubName();
                if (cname.equals(name)) {
                    return ftc;
                }
            }
        } catch (Exception e) {
            System.out.println("Integers are not allowed");
        }
    }
    return null;
}
```

### *2.4.2.4) Displaying Premier League Table*

```
/**
* display statistics of all registered teams
*/
@Override
public void displayLeague() {
    System.out.println();
    if (footballClubList.isEmpty()) {
        System.out.println("No Club Registered To League!");
    } else {
        footballClubList.sort(new FootballClubSort());
        System.out.println("+--------------------+------------------------+---------------+-------------
-+----------------+----------------------+------------------------+-------------------------+--------------
+");
        System.out.println("|     Club Name     |   Matches Played(MP)   |   Wins(W)   |
Draws(D)  |   Defeats(L)   |   Goals Scored(GS)   |   Goals Received(GR)   |   Goals
Difference(GD)  |   Points(P)  |");
        System.out.println("+--------------------+------------------------+---------------+-------------
-+----------------+----------------------+------------------------+-------------------------+--------------
+");
        String leftAlignFormat = "| %-19s |         %-13d |     %-8d |      %-6d |       %-8d |
%-11d |          %-12d |          %-13d |       %-6d |%n";
        for (FootballClub fc : footballClubList) {
            System.out.format(leftAlignFormat,
                fc.getClubName(),
```

```
                fc.getTotalMatches(),
                fc.getWins(),
                fc.getDraws(),
                fc.getDefeats(),
                fc.getGoalsScored(),
                fc.getGoalsReceived(),
                fc.getGoalsDifference(),
                fc.getPoints());
        }
        System.out.println("-------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------
----------\n");
    }
  }
```

### *2.4.2.5) Adding a played match (Set a match)*

```
   /**
    * add match
   * @param teamname1 name as home team
   * @param score1 home team score
   * @param teamname2 club name as away team
   * @param score2 away team score
   * @param date date match
   * @return Match Object
   */
   private Match prepareMatch(String teamname1, int score1, String teamname2, int
score2, String date) {
      int i = 1;
      FootballClub homeClub;
      FootballClub awayClub;
      if (footballClubList.isEmpty()) {
         System.out.println("\n" + teamname1 + " isn't registered In League\n Registering
Football Club \""+teamname1 +"\" to the League ");
         homeClub = createNewClub();
         addFootballClub(homeClub);
         System.out.println("\n" + teamname2 + " isn't registered In League\n Registering
Football Club \""+teamname2 +"\" to the League ");
         awayClub = createNewClub();
         addFootballClub(awayClub);
      }
      for (FootballClub ft1 : footballClubList) {
         if (ft1.getClubName().equals(teamname1)) {
            if (score1 > score2) {
               ft1.setWins(1);
            } else if (score1 < score2) {
               ft1.setDefeats(1);
```

```
            }
            ft1.updateScore(score1, score2);
            break;
          } else if (i == footballClubList.size()) {
            if (this.footballClubList.size() >= AppData.MAX_TEAMS){
            System.out.println("All the slots have been filled");
            return null;
           }
            System.out.println("\n" + teamname1 + " isn't registered In League\n
Registering Football Club \""+teamname1 +"\" to the League ");
            homeClub = createNewClub();
            if (score1 > score2) {
               homeClub.setWins(1);

            } else if (score1 < score2) {
               homeClub.setDefeats(1);
            }
            homeClub.updateScore(score1, score2);
            addFootballClub(homeClub);
            break;
          }
          ++i;
       }
       i = 1;
       for (FootballClub ft2 : footballClubList) {
          if (ft2.getClubName().equals(teamname2)) {
            if (score1 < score2) {
               ft2.setWins(1);
            } else if (score1 > score2) {
               ft2.setDefeats(1);
            }
            ft2.updateScore(score2, score1);
            break;
          } else if (i == footballClubList.size()) {
                     if (this.footballClubList.size() >= AppData.MAX_TEAMS){
                    System.out.println("All the slots have been filled");
                    return null;
           }
            System.out.println("\n" + teamname2 + " isn't registered in League\n
Registering Football Club \""+teamname2 +"\" to the League ");
            awayClub = createNewClub();
            if (score1 < score2) {
               awayClub.setWins(1);
            }else if (score1 > score2) {
               awayClub.setDefeats(1);
            }
            awayClub.updateScore(score2, score1);
            addFootballClub(awayClub);
```

```
            break;
        }
        ++i;
    }

    AppData.modifiedCount += 1;

    Match match = new Match();
    String formatDate = PremierLeagueUtils.formatDate(date);
    match.setDatePlayed(formatDate);
    match.setHomeTeam(teamname1);
    match.setAwayTeam(teamname2);
    match.setHomeScore(score1);
    match.setAwayScore(score2);
    matchData.add(match);
    return match;
  }



   /**
  * form to entry new club
  */
  private FootballClub createNewClub() {
     try {
        FootballClub fc = new FootballClub();
        String name, location, manager;
        int yearFormed = 1900;
        name = keyScanner.inputWithValidation("Enter Club Name  : ", ">>> Club Name
cannot be empty");

        location = keyScanner.inputWithValidation("Enter Club Location  : ", ">>> Club
location cannot be empty");

        manager = keyScanner.inputWithValidation("Enter Club manager  : ", ">>> Club
manager cannot be empty");

        boolean inputValid = false;
        System.out.print("Enter Formed year (Founded year) of the Club: ");
        while (!inputValid) {
           try {
              String input = keyScanner.readLine();
              yearFormed = Integer.parseInt(input);
              inputValid = true;
           } catch (NumberFormatException e) {
              System.out.println("Only integers are allowed...Please re-enter Formed year
(Founded year) of the Club again: ");
```

```
            }
        }

        fc.setClubName(name.trim());
        fc.setClubManager(manager.trim());
        fc.setClubFormedYear(yearFormed);
        fc.setClubLocation(location.trim());
        return fc;
      } catch (IOException e) {
        System.out.println(e.toString());
        return null;
      } catch (Exception e) {
        System.out.println(e.toString());
        return null;
      }
    }



    /**
    * add match
    * @param homeTeamName club name as home team
    * @param homeScore home team score
    * @param awayTeamName club name as away team
    * @param awayScore away team score
    * @param dateFormat date match
    * @return Match Object
    */
    @Override
    public Match footBallMatch(String homeTeamName, int homeScore, String
awayTeamName, int awayScore, String dateFormat) {
        return prepareMatch(homeTeamName, homeScore, awayTeamName, awayScore,
dateFormat);
    }
```

### *2.4.2.5) Display All Matches Played*

```
/**
* Displays All Matches Played with date
*/
    @Override
    public void displayMatchesPlayed() {
        System.out.println("\nDiplaying All Played Matches\n");
        if (matchData.isEmpty()) {
            System.out.println("No Matches Played !");
```

```
    } else {
        for (Match mt : matchData) {
            System.out.printf("\nDate: %s\n%s    vs    %s\n%s     :     %s ",
mt.getDatePlayed(), mt.getHomeTeam(), mt.getAwayTeam(), mt.getHomeScore(),
mt.getAwayScore());
            System.out.println();
        }
    }
    System.out.println();
  }
```

### *2.4.2.6) Save Data to file*

```
  /**
   * save all football club data from ArrayList and put it into txt file
   * @param fileName file contains football club data
   */
  @Override
  public void saveStatisticsData(String fileName) {
    if (AppData.modifiedCount > 0) {
      try {
        FileOutputStream fileOutputStream = new FileOutputStream(fileName);
        ObjectOutputStream objectOutputStream = new
ObjectOutputStream(fileOutputStream);

        for (FootballClub footballClub : footballClubList) {
          objectOutputStream.writeObject(footballClub);
        }

        objectOutputStream.flush();
        fileOutputStream.close();
        objectOutputStream.close();
        System.out.println("Statistics Data has been saved successfully");
      } catch (IOException e) {
        System.out.println("No statistics data present to save in file");
      }
    }else{
      System.out.println("No statistics data present to save in file");
    }

  }
```



```
  /**
```

```
    * save all match data from ArrayList and put it into txt file
    * @param fileName file contains match data
    */
    @Override
    public void saveMatchesData(String fileName) {
        if (AppData.modifiedCount > 0) {
            try {
                FileOutputStream fileOutputStream = new FileOutputStream(fileName);
                ObjectOutputStream objectOutputStream = new
ObjectOutputStream(fileOutputStream);

                for (Match match : matchData) {
                    objectOutputStream.writeObject(match);
                }

                objectOutputStream.flush();
                fileOutputStream.close();
                objectOutputStream.close();
                System.out.println("Matches Data has been saved successfully");
            } catch (IOException e) {
                System.out.println("No matches played data present to save in file");
            }
        }else{
            System.out.println("No matches played data present to save in file");
        }

    }
```

### *2.4.2.7) Load Data from File*

```
     /**
    * load all football club data from file and put it into ArrayList
    * @param fileName file contains footballclub data
    */
    @Override
    public void retrieveStatisticsData(String fileName) {
        footballClubList.clear();
        try {
            FileInputStream fileInputStream = new FileInputStream(fileName);
            ObjectInputStream objectInputStream = new
ObjectInputStream(fileInputStream);
            for (; ; ) {
                try {
                    Object readObject = objectInputStream.readObject();
                    if (readObject instanceof FootballClub) {
                        footballClubList.add((FootballClub) readObject);
                    }
```

```
            } catch (EOFException e) {
                break;
            }
        }
        fileInputStream.close();
        objectInputStream.close();
        System.out.println("Statistics Data has been loaded successfully");
    } catch (IOException | ClassNotFoundException e) {
        System.out.println("No saved records present to load from file");
    }
}



/**
* load all match data from file and put it into ArrayList
* @param fileName file contains match data
*/
@Override
public void retrieveMatchesData(String fileName) {
    matchData.clear();
    try {
        FileInputStream fileInputStream = new FileInputStream(fileName);
        ObjectInputStream objectInputStream = new
ObjectInputStream(fileInputStream);
        for (; ; ) {
            try {
                Object readObject = objectInputStream.readObject();
                if (readObject instanceof Match) {
                    matchData.add((Match) readObject);
                }
            } catch (EOFException e) {
                break;
            }
        }
        fileInputStream.close();
        objectInputStream.close();
        System.out.println("Matches played Data has been loaded successfully");
    } catch (IOException | ClassNotFoundException e) {
        System.out.println("No saved records present to load from file");
    }
}
```

#### *2.4.2.8) View Premier League GUI*

```
/**
* Open browser as app GUI automatically with address host:4200
*/
@Override
public void openWebGUI() {
   System.out.println("Please Wait...Opening Premier League GUI");
   String url = "http://localhost:4200";
   String os = System.getProperty("os.name").toLowerCase();
   Runtime rt = Runtime.getRuntime();
   try {
      if (os.contains("win")) {   //for windows
         rt.exec("rundll32 url.dll,FileProtocolHandler " + url);

      } else if (os.contains("mac")) {   //for mac os
         rt.exec("open " + url);

      } else if (os.contains("nix") || os.contains("nux")) {   // for linux
         rt.exec("xdg-open " + url);
      }
   } catch (Exception ignored) {
   }
}
```

### 2.4.3)PremierLeagueApplication class

```
package cli;

import entities.match.Date;
import entities.team.FootballClub;
import utils.KeyScanner;
import data.AppData;

import java.util.Objects;
import java.io.IOException;

/**
* Class to run CLI Application
*
*/
public class PremierLeagueApplication {
   private LeagueManager manager = new PremierLeagueManager();
   private KeyScanner keyScanner = new KeyScanner();

   /**
   * Run console app
   * @param leagueFile path file / filename that store Football Club data
```

```
   * @param matchFile path file / filename that store Match data
   */
   public void runConsole(String leagueFile, String matchFile) throws IOException {
      manager.retrieveStatisticsData(leagueFile);
      manager.retrieveMatchesData(matchFile);

System.out.println("============================================================
==============================\n");
      System.out.println("**~**********~**********~** - WELCOME TO PREMIER LEAGUE
APPLICATION -  *~**********~**********~**");


      boolean showMenu = true;
      while (showMenu) {
         displayMenu();

         String choice = keyScanner.readLine();

         switch (choice) {
            case "1":
               addFootballClub();
               break;
            case "2":
               deleteClub();
               break;
            case "3":
               displayStatistics();
               break;
            case "4":
               displayLeague();
               break;
            case "5":
               setMatches();
               break;
            case "6":
               displayMatchesPlayed();
               break;
            case "7":
               manager.saveStatisticsData(leagueFile);
               manager.saveMatchesData(matchFile);
               AppData.modifiedCount = 0;
               break;
            case "8":
               manager.openWebGUI();
               break;
            case "0":
               showMenu = false;
               System.out.println("Thank you for using the system.....");
```

```
                break ;
            default:
                System.out.println("Invalid option entered!!!....Please re-enter");
                break;
        }
    }
  }

  /**
  * Display menu option for Premier League CLI App.
  */
  private void displayMenu(){

System.out.println("\n==========================================================
==================================\n");
      System.out.println("Enter (1) - Register (Create) A Club");
      System.out.println("Enter (2) - Delete A Club");
      System.out.println("Enter (3) - Display Statistics of a Selected Club ");
      System.out.println("Enter (4) - Display PremierLeague Table");
      System.out.println("Enter (5) - Add a Played Match with date (Set a Match) ");
System.out.println("Enter (6) - Display All Matches Played ");
      System.out.println("Enter (7) - Save Data to File ");
      System.out.println("Enter (8) - Open Premier League Web GUI ");
      System.out.println("Enter (0) - Quit");

System.out.println("==========================================================
==============================\n");
      System.out.println("> Please Select An Option From The Following");
  }



  /**
  * Add football club
  */
  private void addFootballClub() throws IOException{
      FootballClub footballClub;
      String clubName = keyScanner.inputWithValidation("\nEnter Club  : ", ">>> Club
cannot be empty");

      String clubLoc = keyScanner.inputWithValidation("\nEnter Club Location: ", ">>>
Club location cannot be empty");

      String managerName = keyScanner.inputWithValidation("\nEnter name of the Club
Manager: ", ">>> Club manager cannot be empty");
```



```
      boolean inputValid = false;
      System.out.print("\nEnter Formed year (Founded year) of the Club: ");
```

```
      while (!inputValid) {
         try {
            String input = keyScanner.readLine();
            int formedYear = Integer.parseInt(input);
            inputValid = true;

            footballClub = new FootballClub(clubName.trim(), clubLoc.trim(),
managerName.trim(), formedYear);
            manager.addFootballClub(footballClub);
         } catch (NumberFormatException e) {
            System.out.println("Only integers are allowed...Please re-enter Formed year
(Founded year) of the Club again: ");
         }
      }


   }

   /**
   * Delete football club
   */
   private void deleteClub() throws IOException {
      String name = keyScanner.inputWithValidation("\nPlease enter club name you
want to delete : ", ">>> Club name cannot be empty");
      manager.deleteClub(name.trim());
   }

   /**
   * Display football club statistics
   */
   private void displayStatistics() {
      boolean inputValid = false;
      System.out.println("\nEnter Name of Club to display statistics: ");
      while (!inputValid) {
         try {
            String name = keyScanner.inputWithValidation("Enter Club Name  : ", ">>>
Club Name cannot be empty");
            FootballClub ftc = manager.findTeamStatisticsByName(name.trim());
            if (Objects.isNull(ftc)) {
               System.out.println("No such club exists");
               break;
            } else {
               System.out.printf("\nClub Name: %s\nClub Location: %s\nClub Manager:
%s\nClub Formed Year: %s\nMatches played: %s\nWins: %s\nDraws: %s\nDefeat:
%s\nGoals Scored: %s\nGoals Received: %s\nGoals Difference: %s\nPoints: %s\n",
ftc.getClubName(), ftc.getClubLocation(),ftc.getClubManager(),
ftc.getClubFormedYear(),ftc.getTotalMatches(), ftc.getWins(), ftc.getDraws(),
ftc.getDefeats(), ftc.getGoalsScored(), ftc.getGoalsReceived(), ftc.getGoalsDifference(),
```

```
ftc.getPoints());
                System.out.println();
                inputValid = true;
            }

        } catch (Exception e) {
            System.out.println("No integers are allowed...Please re-enter name of the club
to display statistics ");
        }
    }

  }

  /**
  * Set Match by entry home team and away team. If teams that are entered is not
registered,
  * App will open prompt to add football club
  */
  private void setMatches() throws IOException {
    Date date = new Date();
    System.out.println("\nSetting Matches...\n");
    String homeTeamName, awayTeamName;
    int homeScore = -1, awayScore = -1;
    System.out.print("Enter Date Of Match(DD/MM/YYY)");
    int year, month, day;
    String dateFormat = "";
    boolean inputValidDay = false, inputValidMonth = false, inputValidYear = false,
        inputValidHomeScore = false, inputValidAwayScore = false;

    while (!inputValidDay) {
      try {
        System.out.print("\nDay: ");
        String dd = keyScanner.readLine();
        day = Integer.parseInt(dd);
        if (date.isValidDay(day)) {
          dateFormat += "" + day;
          inputValidDay = true;
        } else {
          System.out.println("Invalid Day Input!");
        }
      } catch (NumberFormatException e) {
        System.out.println("Invalid Day Input!");
      }
    }
```



```
    while (!inputValidMonth) {
      try {
```

```
            System.out.print("Month: ");
            String mm = keyScanner.readLine();

            month = Integer.parseInt(mm);
            if (date.isValidMonth(month)) {
               dateFormat += "/" + month + "/";
               inputValidMonth = true;
            } else {
               System.out.println("Invalid Month Input!");
            }
         } catch (NumberFormatException e) {
            System.out.println("Invalid Month Input!");
         }
      }

      while (!inputValidYear) {
         try {
            System.out.print("Year: ");
            String yy = keyScanner.readLine();

            year = Integer.parseInt(yy);
            if (date.isValidYear(year)) {
               dateFormat += "" + year;
               inputValidYear = true;
            } else {
               System.out.println("Invalid Year Input!");
            }
         } catch (NumberFormatException e) {
            System.out.println("Invalid Year Input!");
         }
      }

      System.out.println();

      System.out.print("Home Team Setup\n");
      homeTeamName = keyScanner.inputWithValidation("Enter Club Name  : ", ">>>
Club Name cannot be empty");

      while (!inputValidHomeScore) {
         try {
            System.out.print("Enter Club Score: ");
            String scoreHome = keyScanner.readLine();

            homeScore = Integer.parseInt(scoreHome);
            if (homeScore >= 0) {
               inputValidHomeScore = true;
            } else {
               System.out.println("Invalid Score Input!");
```

```
            }
        } catch (NumberFormatException e) {
            System.out.println("Invalid Score Input!");
        }
    }

    System.out.println();

    System.out.print("Away Team Setup\n");
    awayTeamName = keyScanner.inputWithValidation("Enter Club Name  : ", ">>>
Club Name cannot be empty");

    while (!inputValidAwayScore) {
        try {
            System.out.print("Enter Club Score: ");
            String scoreAway = keyScanner.readLine();

            awayScore = Integer.parseInt(scoreAway);
            if (awayScore >= 0) {
                inputValidAwayScore = true;
            } else {
                System.out.println("Invalid Score Input!");
            }
        } catch (NumberFormatException e) {
            System.out.println("Invalid Score Input!");
        }
    }

    System.out.println("\nMatch has been added successfully");
    manager.footBallMatch(homeTeamName.trim(), homeScore,
awayTeamName.trim(), awayScore, dateFormat);
  }

  /**
  * Display football club list with their statistics
  */
  private void displayLeague()  {
    manager.displayLeague();
  }

  /**
  * Display all match
  */
  private void displayMatchesPlayed(){
    manager.displayMatchesPlayed();
  }
}
```

## 2.5) services

### 2.5.1) MatchService Interface

```
package services.match;

import com.google.inject.ImplementedBy;

import entities.match.Match;

import java.util.List;

@ImplementedBy(MatchServiceImpl.class)
public interface MatchService {
   List<Match> getAllMatches();

   List<Match> getMatchesByDate(String date);

   List<Match> getMatchesByDateAsc();

   Match randomMatch();

   void save(Match match);
}
```

### 2.5.2) MatchServiceImpl class

```
package services.match;

import entities.match.Match;
import entities.team.FootballClub;
import utils.sort.DateSort;
import utils.PremierLeagueUtils;
import data.AppData;
import services.team.TeamService;

import java.io.*;
import java.util.ArrayList;
import java.util.List;
import javax.inject.Inject;
import com.typesafe.config.Config;

/**
```

```
* Class that handles Match data
*/
public class MatchServiceImpl implements MatchService {
   private final Config config;
   private final TeamService teamService;

   @Inject
   public MatchServiceImpl(Config config, TeamService teamService) {
      this.config = config;
      this.teamService = teamService;
   }

   /**
   * get all match data
   * @return {@code List<MatchData>}
   */
   @Override
   public List<Match> getAllMatches() {
      return AppData.matchData;
   }

   /**
   * get match data by date
   * @param date
   * @return {@code List<MatchData>}
   */
   @Override
   public List<Match> getMatchesByDate(String date) {
      List<Match> allMatches = this.getAllMatches();
      List<Match> allMatchesByDate = new ArrayList<>();
      for (Match match : allMatches) {
         String formatDate = PremierLeagueUtils.formatDate(date);
         if (formatDate.equals(match.getDatePlayed())) {
            allMatchesByDate.add(match);
         }
      }
      return allMatchesByDate;
   }

   /**
   * get all match data order by date ascending
   * @return {@code List<MatchData>}
   */
   @Override
   public List<Match> getMatchesByDateAsc() {
      List<Match> all = this.getAllMatches();
      all.sort(new DateSort());
      return all;
```

```
    }

    /**
    * generate random Match
    * @return {@code MatchData}
    */
    @Override
    public Match randomMatch() {
        FootballClub home = getRandomFootballClub();
        FootballClub away = getRandomFootballClub();
        while (home.getClubName().equals(away.getClubName())) {
            away = getRandomFootballClub();
        }
        return generateMatch(home, away);
    }

    /**
    * generate random Match process
    * @return {@code MatchData}
    */
    private Match generateMatch(FootballClub home, FootballClub away) {
        int goalsScoredHome = PremierLeagueUtils.generateRandomNumber(6);
        int goalsScoredAway = PremierLeagueUtils.generateRandomNumber(6);
        if (goalsScoredHome < goalsScoredAway) {
            // Away Win
            home.setDefeats(1);
            away.setWins(1);
            away.updateScore(goalsScoredAway, goalsScoredHome);
            home.updateScore(goalsScoredHome, goalsScoredAway);

        } else if (goalsScoredHome > goalsScoredAway) {
            // Home Win
            away.setDefeats(1);
            home.setWins(1);
            home.updateScore(goalsScoredHome, goalsScoredAway);
            away.updateScore(goalsScoredAway, goalsScoredHome);
        }
                else {
            // Match Draw
            home.updateScore(goalsScoredHome, goalsScoredAway);
            away.updateScore(goalsScoredAway, goalsScoredHome);
        }
        Match match = new Match(PremierLeagueUtils.generateNow(),
home.getClubName(), away.getClubName(), goalsScoredHome, goalsScoredAway);
        this.save(match);
        teamService.save(home);
        teamService.save(away);
        return match;
```

```
    }

    /**
    * generate random football club
    * @return {@code FootballClub}
    */
    private FootballClub getRandomFootballClub() {
        List<FootballClub> footballClubList = teamService.getFootballLeague();
        return
footballClubList.get(PremierLeagueUtils.generateRandomNumber(footballClubList.size(
)));
    }

    /**
    * save all match data from ArrayList and put it into txt file
    * @param match file contains match data
    */
    @Override
    public void save(Match match) {
        List<Match> allMatches = this.getAllMatches();
        allMatches.add(match);
        try {
            String matchFile = config.getString("file.match_file_name");
            FileOutputStream fileOutputStream = new FileOutputStream(matchFile);
            ObjectOutputStream objectOutputStream = new
ObjectOutputStream(fileOutputStream);

            allMatches.forEach(el -> {
                try {
                    objectOutputStream.writeObject(el);
                } catch (IOException e) {
                    e.printStackTrace();
                }
            });
            fileOutputStream.close();
            objectOutputStream.close();

        } catch (IOException e) {
            e.printStackTrace();
        }
    }

}
```

### 2.5.3) TeamService Interface

```
package services.team;

import com.google.inject.ImplementedBy;

import entities.team.FootballClub;

import java.util.List;

@ImplementedBy(TeamServiceImpl.class)
public interface TeamService {
   List<FootballClub> getFootballLeague();

   List<FootballClub> getFootballLeagueOrder(String orderBy);

   void save(FootballClub footballClub);
}
```

### 2.5.4) TeamServiceImpl class

```
package services.team;

import entities.team.FootballClub;
import utils.sort.FootballClubSort;
import utils.sort.GoalsSort;
import utils.sort.WinSort;
import data.AppData;
import java.io.*;
import java.util.List;
import java.util.ListIterator;
import javax.inject.Inject;
import com.typesafe.config.Config;

/**
* Class that handles Football Club data
*/
public class TeamServiceImpl implements TeamService {

   private final Config config;

   @Inject
   public TeamServiceImpl(Config config) {
      this.config = config;
```

```
    }

    /**
    * get all football club
    * @return {@code List<FootballClub>}
    */
    @Override
    public List<FootballClub> getFootballLeague() {
        return AppData.footballClubList;
    }

    /**
    * get football club order by option descending
    * @param orderBy sort option (goals, wins, points)
    * @return {@code List<FootballClub>}
    */
    @Override
    public List<FootballClub> getFootballLeagueOrder(String orderBy) {
        List<FootballClub> footballClubList = this.getFootballLeague();
        switch (orderBy) {
            case "goals":
                footballClubList.sort(new GoalsSort());
                break;
            case "wins":
                footballClubList.sort(new WinSort());
                break;
            case "points":
                footballClubList.sort(new FootballClubSort());
                break;
        }
        return footballClubList;
    }

    /**
    * save all football club data from ArrayList and put it into txt file
    * @param footballClub file contains football club data
    */
    @Override
    public void save(FootballClub footballClub) {
        List<FootballClub> allTeams = this.getFootballLeague();
        ListIterator<FootballClub> footballClubListIterator = allTeams.listIterator();
        while (footballClubListIterator.hasNext()) {
            if
(footballClubListIterator.next().getClubName().equals(footballClub.getClubName())) {
                footballClubListIterator.set(footballClub);
            }
        }
        try {
```

```
            String leagueFile = config.getString("file.league_file_name");
            FileOutputStream fileOutputStream = new FileOutputStream(leagueFile);
            ObjectOutputStream objectOutputStream = new
    ObjectOutputStream(fileOutputStream);

            allTeams.forEach(el -> {
               try {
                  objectOutputStream.writeObject(el);
               } catch (IOException e) {
                  e.printStackTrace();
               }
            });
            fileOutputStream.close();
            objectOutputStream.close();

         } catch (IOException e) {
            e.printStackTrace();
         }
      }

    }
```

## 2.6) controllers

### 2.6.1)MatchController class

```
/**
* Cannot use "match" keyword because in Scala (the main language that builds the play
framework Play Framework)
* "match" keyword is reverse keyword
*
*/

package controllers.matches;

import javax.inject.Inject;

import play.mvc.*;
import play.libs.Json;

import services.match.MatchService;
import entities.match.Match;
import utils.PremierLeagueUtils;

import java.util.List;
```



```
/**
* Class that handle MatchesPlayed API data
```

```
*/
public class MatchController extends Controller {
        private final MatchService matchService;

        @Inject
        public MatchController(MatchService matchService){
                this.matchService = matchService;
        }

  /**
   * To Find a match by date.The result is {@code List<Match>} with all matches in the
given date {@code String date}
   * path = "/matches"
   *
   * @param date (default value = "") date parameter to get matches
   * @return {@code List<Match>} or ResponseEntity no content in case of no results
   *
   */
  public Result findMatchesByDate(String date) {
    if (!date.isEmpty() && !PremierLeagueUtils.checkValidDate(date)) {
      return status(409, "Malformed date");
    }
    try {
      List<Match> matches = (date.isEmpty()) ? matchService.getAllMatches() :
matchService.getMatchesByDate(date);
      if (matches.size() > 0) {
        return ok(Json.toJson(matches));
      } else {
        System.out.println("no content");
        return noContent();
      }
    } catch (Exception ex) {
      ex.printStackTrace();
      return badRequest();
    }
  }

  /**
   * Gets all matches in ascending order by date. The result is {@code List<Match>}
with all matches order by date
   * path = "/matches/order-by-date"
   *
   * @return {@code List<Match>} order descending date or ResponseEntity no
content in case of no results.
   */
  public Result getMatchesOrderAscByDate() {
    try {
      List<Match> matches = matchService.getMatchesByDateAsc();
```

```
            if (matches.size() > 0) {
                return ok(Json.toJson(matches));
            } else {
                return noContent();
            }
        } catch (Exception ex) {
            ex.printStackTrace();
            return badRequest();
        }
    }


     /**
     * Start a match between two randoms {@code FootballClub} teams the result is a
{@code Match}
     * otherwise a bad request is returned.
     * path = "/random"
     *
     * @return Random {@code Match} or ResponseEntity bad request.
     */
    public Result generateRandomMatch() {
        Match randomMatch = matchService.randomMatch();
        try {
            return ok(Json.toJson(randomMatch));
        } catch (Exception ex) {
            return badRequest();
        }
    }}
```

### 2.6.2)TeamController class

```
package controllers.team;

import javax.inject.Inject;

import play.mvc.*;
import play.libs.Json;

import services.team.TeamService;
import entities.team.FootballClub;

import java.util.List;
```



```
/**
* Class that handle FootballClub API data
*/
```

```
public class TeamController extends Controller {
        private final TeamService teamService;

        @Inject
        public TeamController(TeamService teamService){
                this.teamService = teamService;
        }

   /**
    * Get unsorted list of FootballClub teams registered in Premier League by
default.The result is {@code List<FootballClub>} or HTTP no content in case of no
results , if want to get the list sorted by wins,goals or points need to send  {@code
String orderBy} parameter in the request.
    * @param orderBy (default value = null) The string to get sorted list (points,wins or
goals)
    * @return {@code List<FootballClub>} or HTTP no content in case of no results
    */
   public Result getTeams(String orderBy) {
      try{
         List<FootballClub> footballClubList = (orderBy.isEmpty()) ?
teamService.getFootballLeague() : teamService.getFootballLeagueOrder(orderBy);
         if (footballClubList.size() > 0) {
            return ok(Json.toJson(footballClubList));
         } else {
            return noContent();
         }
      }catch(Exception ex) {
         return badRequest();
      }

   }

}
```

### 2.6.3)FrontendController class

```
package controllers;

import play.mvc.*;

import javax.inject.Inject;
import com.typesafe.config.Config;
import java.io.File;
```



```
/**
* Class that handle Angular UI app, act as a proxy that allow Play App to access all
```

```
resources or file in Angular UI Project
*/
public class FrontendController extends Controller {
   private final Config config;

   @Inject
   public FrontendController(Config config){
    this.config = config;
   }

   /**
   * handle access of UI index page
   */
   public Result index() {
    File file = new File("ui/src/index.html");
    return ok(file, true);
   }

   /**
   * Handle all UI Project resources so Play App can access it from its port
   */
   public Result assetOrDefault(String resource){
    if(resource.startsWith(config.getString("apiPrefix"))){
      return notFound();
    }else{
      if (resource.contains(".")){
        File file = new File("ui/src/"+resource);
        return ok(file, true);
      }else{
        return index();
      }
    }

   }

}
```

## 2.7) tasks

### 2.7.1) AppThread class

```
package tasks;

import cli.PremierLeagueApplication;
import javax.inject.Inject;
import com.typesafe.config.Config;
import java.io.IOException;

/**
* This class is responsible to handle thread that make CLI application to run in Play
Framework console.
* AppThread extends Java Thread which allows this process to run in background not in
foreground
*/
public class AppThread extends Thread{
        private final Config config;
        private final PremierLeagueApplication premierLeagueApp;

        /**
        * Constructor of AppThread
        * @param premierLeagueApp instance of PremierLeagueApplication class
        * @param config The Class which allows to access data from configuration file
like application.conf
        */
        @Inject
        public AppThread(PremierLeagueApplication premierLeagueApp, Config config){
                this.premierLeagueApp = premierLeagueApp;
                this.config = config;
        }

        /**
        * method that called when this class instantiated or loaded
        */
        @Override
   public void run(){
        System.out.println("Console App Running");

      String leagueFile = config.getString("file.league_file_name");
      String matchFile = config.getString("file.match_file_name");

      try{
         premierLeagueApp.runConsole(leagueFile, matchFile);
      }catch(IOException e){
         System.out.println(e.toString());
```

```
        }

    }

}
```

### 2.7.2) InitActorTask class

```
package tasks;

import entities.team.FootballClub;
import entities.match.Match;
import data.AppData;
import cli.PremierLeagueManager;

import java.util.List;
import javax.inject.Inject;
import com.typesafe.config.Config;

import play.DefaultApplication;

public class InitActorTask {
   private final Config config;
   private final AppThread appThread;
   private final PremierLeagueManager premierLeagueManager;

   private List<FootballClub> footballClubList = AppData.footballClubList;
   private List<Match> matchData = AppData.matchData;

   private final DefaultApplication defaultApp;

   /**
   * The constructor of InitActorTask Class
   * this method calls initialize method and start CLI application Thread
   */
   @Inject
   public InitActorTask(Config config, AppThread appThread, PremierLeagueManager
premierLeagueManager, DefaultApplication defaultApp){
      this.config = config;
      this.appThread = appThread;
      this.premierLeagueManager = premierLeagueManager;
      this.defaultApp = defaultApp;
      this.initialize();

      if(!defaultApp.isTest()){
         appThread.start();
      }
   }
```

```
    /**
    * this method load footballclub(league) data and match data from txt file
    */
    private void initialize(){
       System.out.println("Load Inital Data from File");

       String leagueFile = config.getString("file.league_file_name");
       String matchFile = config.getString("file.match_file_name");

       premierLeagueManager.retrieveStatisticsData(leagueFile);
       premierLeagueManager.retrieveMatchesData(matchFile);
    }
}
```

### 2.7.3) InitTasksModule class

```
package tasks;

import com.google.inject.AbstractModule;


/**
* module which allows to run command in startup of Play Framework
*/
public class InitTasksModule extends AbstractModule {

        /**
        * this function gets called when this class is loaded
        */
        @Override
        protected void configure() {
        bind(InitActorTask.class).asEagerSingleton();
        }

}
```

# 3) Unit Testing

## 3.1) TeamTest class

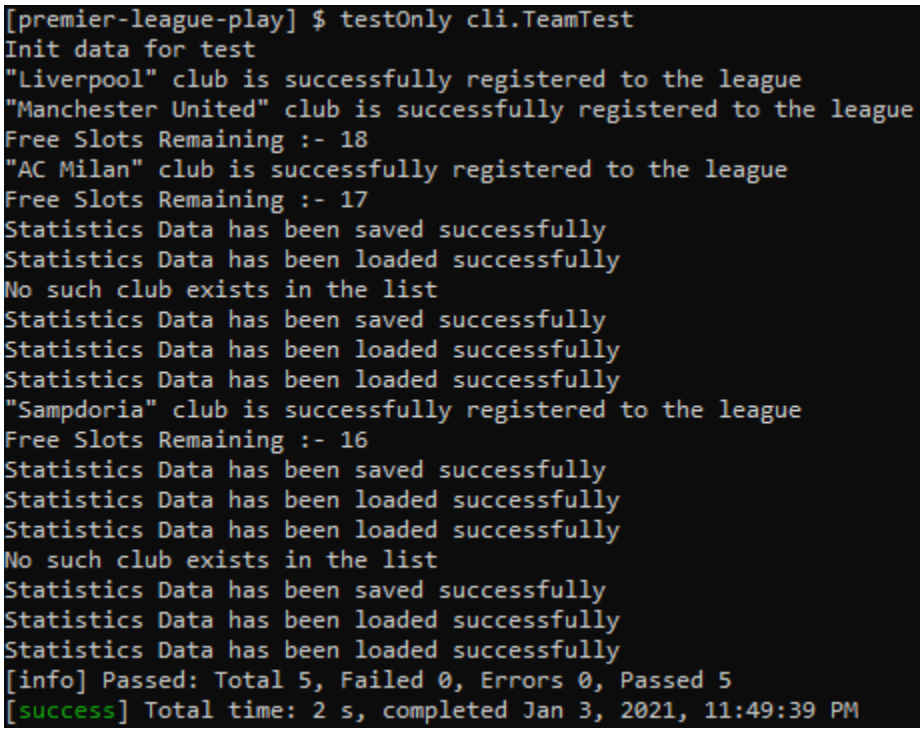
- This Class is used to ensure Team Feature of CLI part running well

```
package cli;

import cli.LeagueManager;
import cli.PremierLeagueManager;
import entities.team.FootballClub;
import org.junit.BeforeClass;
import org.junit.Test;

public class TeamTest {
   // this is to work with one instance of PremierLeagueManager
   private static LeagueManager premierLeague() {
      return new PremierLeagueManager();
   }

   /**
    * Execute initTestData before all test
    */

   @BeforeClass
   public static void initTeamTestData() {
      System.out.println("Init data for test");
      FootballClub liverpoolTeam = new FootballClub();
      liverpoolTeam.setClubName("Liverpool");
      liverpoolTeam.setClubLocation("England");
      liverpoolTeam.setClubManager("William");
      liverpoolTeam.setClubFormedYear(1978);

      FootballClub manchesterTeam = new FootballClub();
      manchesterTeam.setClubName("Manchester United");
      manchesterTeam.setClubLocation("England");
      manchesterTeam.setClubManager("Ben");
      manchesterTeam.setClubFormedYear(1978);

      FootballClub milanTeam = new FootballClub();
      milanTeam.setClubName("AC Milan");
      milanTeam.setClubLocation("Italy");
      milanTeam.setClubManager("Carletto");
      milanTeam.setClubFormedYear(1899);
```

```
        premierLeague().addFootballClub(liverpoolTeam);
        premierLeague().addFootballClub(manchesterTeam);
        premierLeague().addFootballClub(milanTeam);

        premierLeague().saveStatisticsData("LeagueFileTest.txt");
    }

    /**
    * add football club test
    */
    @Test
    public void addFootballClubTest() {
        boolean findNewClub = false;
        FootballClub footballClub = new FootballClub();
        footballClub.setClubName("Sampdoria");
        footballClub.setClubLocation("Italy");
        footballClub.setClubManager("Genaro Gatuso");
        footballClub.setClubFormedYear(1878);

        premierLeague().addFootballClub(footballClub);  // add footballClub to array list

        premierLeague().saveStatisticsData("LeagueFileTest.txt");  // save all data in file

        premierLeague().retrieveStatisticsData("LeagueFileTest.txt");  // retrieve all data to
check if the object really saves in it

        // Loop the array list (getClubs return the football array list from premier League)
        for (FootballClub footballClub1 : premierLeague().getClubs()) {
            if (footballClub1.getClubName().equals(footballClub.getClubName())) {
                findNewClub = true;
                break;
            }
        }
        assert findNewClub;
    }

    /**
    * Delete test
    */
    @Test
    public void deleteExistedFootballClubTest() {
        boolean deleteClub = true;
        premierLeague().retrieveStatisticsData("LeagueFileTest.txt");
        premierLeague().deleteClub("Sampdoria");
        premierLeague().saveStatisticsData("LeagueFileTest.txt");
        premierLeague().retrieveStatisticsData("LeagueFileTest.txt");
        for (FootballClub footballClub : premierLeague().getClubs()) {
            if (footballClub.getClubName().equals("Sampdoria")) {
```

```
            deleteClub = false;
            break;
        }
      }
      assert deleteClub;
   }

   @Test
   public void deleteNotExistedFootballClubTest() {
      boolean deleteClub = false;
      premierLeague().retrieveStatisticsData("LeagueFileTest.txt");
      premierLeague().deleteClub("Unknown Team");
      premierLeague().saveStatisticsData("LeagueFileTest.txt");
      premierLeague().retrieveStatisticsData("LeagueFileTest.txt");

      for (FootballClub footballClub : premierLeague().getClubs()) {
         if (footballClub.getClubName().equals("Unknown Team")) {
            deleteClub = true;
            break;
         }
      }
      assert !deleteClub;
   }

   /**
   * Search Test
   */
   @Test
   public void findTeamStatisticsByNameTest() {
      premierLeague().retrieveStatisticsData("LeagueFileTest.txt");
      FootballClub footballClubTest = premierLeague().findTeamStatisticsByName("AC
Milan");
      assert footballClubTest != null;
   }

   /**
   * Test not existed team by name
   */

   @Test
   public void findNotExistedTeamStatisticsByNameTest() {
      premierLeague().retrieveStatisticsData("LeagueFileTest.txt");
      FootballClub footballClubTest =
premierLeague().findTeamStatisticsByName("Unknown Team");
      assert footballClubTest == null;
   }

}
```

Screenshot of TeamTest class




## 3.2) TeamControllerTest class

- This Class is used to ensure Team Feature of GUI part running well

```
package controllers;

import org.junit.Test;
import play.Application;
import play.inject.guice.GuiceApplicationBuilder;
import play.mvc.Http;
import play.mvc.Result;
import play.test.WithApplication;
```

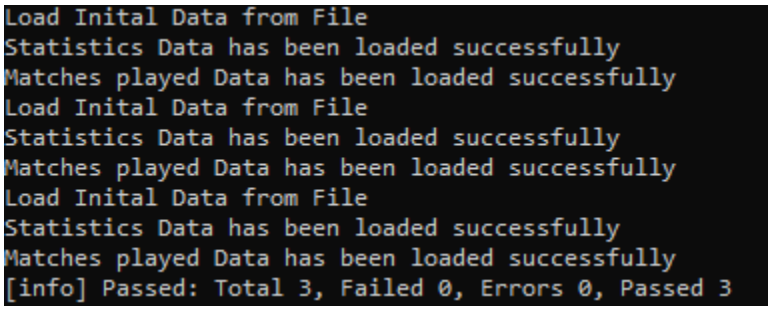
```
import static org.junit.Assert.assertEquals;
import static play.mvc.Http.Status.OK;
import static play.test.Helpers.GET;
import static play.test.Helpers.route;

/**
* This class ensure TeamController running well
* Referred :-
https://www.playframework.com/documentation/2.8.x/JavaFunctionalTest
*/
public class TeamControllerTest extends WithApplication {


  /**
  * Override play app configuration for the test
  */
  @Override
  protected Application provideApplication() {
    return new GuiceApplicationBuilder()
 .configure("file.league_file_name", "LeagueFileTest.txt")
    .configure("file.match_file_name", "MatchFileTest.txt")
 .build();
  }

  /**
  * Test team list order by points, should be http 200 response code
  */
  @Test
  public void getTeamsOrderByPoints() {
    Http.RequestBuilder request = new Http.RequestBuilder()
        .method(GET)
        .uri("http://localhost:9000/premierleague/team/teams?orderBy=points");

    Result result = route(app, request);
    assertEquals(OK, result.status());
  }

  /**
  * Test team list order by goals, should be http 200 response code
  */
      @Test
  public void getTeamsOrderByGoals() {
    Http.RequestBuilder request = new Http.RequestBuilder()
        .method(GET)
        .uri("http://localhost:9000/premierleague/team/teams?orderBy=goals");
```



```
    Result result = route(app, request);
    assertEquals(OK, result.status());
```

```
    }

    /**
    * Test team list order by wins, should be http 200 response code
    */
          @Test
    public void getTeamsOrderByWins() {
       Http.RequestBuilder request = new Http.RequestBuilder()
             .method(GET)
             .uri("http://localhost:9000/premierleague/team/teams?orderBy=wins");

       Result result = route(app, request);
       assertEquals(OK, result.status());
    }
}
```

## Screenshot of TeamControllerTest class




### 3.3) MatchTest class

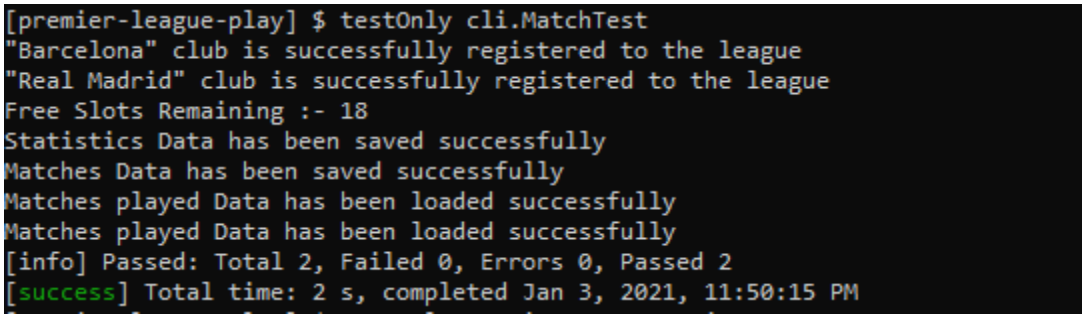
- This Class is used to ensure Match Feature of CLI part running well

```
package cli;

import cli.LeagueManager;
import cli.PremierLeagueManager;
import entities.match.Match;
import entities.team.FootballClub;
import org.junit.BeforeClass;
import org.junit.Test;

/**
```

```
* This Class is used to ensure Match Feature running well
*/
public class MatchTest {
   /**
    * this is to work with one instance of PremierLeagueManager
    */
   private static LeagueManager premierLeague() {
      return new PremierLeagueManager();
   }

   /**
    * Execute initTestData before all test
    */
   @BeforeClass
   public static void initMatchTestData() {
      FootballClub homeClub = new FootballClub();
      homeClub.setClubName("Barcelona");
      homeClub.setClubLocation("Spain");
      homeClub.setClubManager("John");
      homeClub.setClubFormedYear(1899);

      FootballClub awayClub = new FootballClub();
      awayClub.setClubName("Real Madrid");
      awayClub.setClubLocation("Spain");
      awayClub.setClubManager("Peter");
      awayClub.setClubFormedYear(1899);

      premierLeague().addFootballClub(homeClub);
      premierLeague().addFootballClub(awayClub);
      premierLeague().saveStatisticsData("LeagueFileTest.txt");

      premierLeague().footBallMatch(homeClub.getClubName(), 1,
awayClub.getClubName(), 2, "11/12/2020");
      premierLeague().footBallMatch(homeClub.getClubName(), 3,
awayClub.getClubName(), 4, "10/12/2020");
      premierLeague().footBallMatch(homeClub.getClubName(), 2,
awayClub.getClubName(), 1, "28/12/2020");

      premierLeague().saveMatchesData("MatchFileTest.txt");
   }

   @Test
   public void footballMatchDatePlayedTest() {
      premierLeague().retrieveMatchesData("MatchFileTest.txt");
      boolean findMatch = false;
      for (Match match : premierLeague().getMatchData()) {
         if (match.getDatePlayed().equals("28/12/2020")) {
            findMatch = true;
```

```
            break;
          }
        }
        assert findMatch;
    }


    /**
    * test if match not exist
    */
    @Test
    public void footballMatchDatePlayedNotExistTest() {
        premierLeague().retrieveMatchesData("MatchFileTest.txt");
        boolean findMatch = false;
        for (Match match : premierLeague().getMatchData()) {
            if (match.getDatePlayed().equals("10/10/2020")) {
                findMatch = true;
                break;
            }
        }
        assert !findMatch;
    }
}
```

## Screenshot of MatchTest class



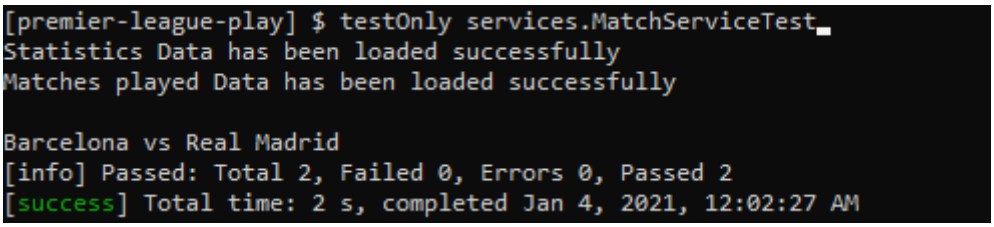
### 3.4) MatchServiceTest class

```
package services;

import org.junit.BeforeClass;
import org.junit.Test;
import org.junit.Assert;

import java.io.File;
import java.util.Date;
import java.util.List;
import java.text.SimpleDateFormat;
import java.text.ParseException;

import entities.match.Match;
import entities.team.FootballClub;
import services.match.*;
import services.team.*;
import cli.PremierLeagueManager;

import com.typesafe.config.Config;
import com.typesafe.config.ConfigFactory;


/**
* This class ensure MatchService Class running well.
* MatchService.class is usefull to deliver match data to controller
*/
public class MatchServiceTest {
	private static MatchService matchService;
	private static TeamService teamService;
	private static PremierLeagueManager premierLeagueManager;


	/**
	* Initial data & object before run test
	*/
	@BeforeClass
	public static void initializeTest(){
		premierLeagueManager = new PremierLeagueManager();
		premierLeagueManager.retrieveStatisticsData("LeagueFileTest.txt");
		premierLeagueManager.retrieveMatchesData("MatchFileTest.txt");

		String configFile = "conf/application_test.conf";
	Config config = ConfigFactory.parseFile(new File(configFile));
	teamService = new TeamServiceImpl(config);
	matchService = new MatchServiceImpl(config, teamService);
	}
```

```
	/**
	* Test randomMatch() method ensure it returns Match data
	*/
	@Test
	public void generateRandomMatchTest(){
		boolean findClub = false;
		Match randomMatch = matchService.randomMatch();

		for (FootballClub footballClub: teamService.getFootballLeague()) {

	if(footballClub.getClubName().equalsIgnoreCase(randomMatch.getHomeTeam()
)){
				findClub = true;
			}
		}

		System.out.println();
		System.out.println(randomMatch.getHomeTeam()+" vs
"+randomMatch.getAwayTeam());
		Assert.assertEquals(true, findClub);
	}


	/**
	* This test ensure match data sorted correctly
	*/
	@Test
	public void sortedMatchDataTest(){
		boolean sorted = false;
		List<Match> sortedMatchs = matchService.getMatchesByDateAsc();
		int totalMatch = sortedMatchs.size();

		Match firstMatch = sortedMatchs.get(0);
		Match lastMatch = sortedMatchs.get(totalMatch - 1);

		SimpleDateFormat format = new SimpleDateFormat("dd/MM/yyyy");

		try{
			Date firstMatchDate = format.parse(firstMatch.getDatePlayed());
			Date lastMatchDate = format.parse(lastMatch.getDatePlayed());
			if (firstMatchDate.before(lastMatchDate)) {
				sorted = true;
			}
		}catch (ParseException pe) {
			System.out.println(pe.toString());
		}catch (Exception e){
			System.out.println(e.toString());
		}
```

```
                Assert.assertEquals(true, sorted);
        }

}
```

Screenshot of MatchServiceTest class




### 3.5) MatchControllerTest class

- This Class is used to ensure Match Feature of GUI part running well

```
package controllers;

import org.junit.Test;
import play.Application;
import play.inject.guice.GuiceApplicationBuilder;
import play.mvc.Http;
import play.mvc.Result;
import play.test.WithApplication;

import static org.junit.Assert.assertEquals;
import static play.mvc.Http.Status.OK;
import static play.test.Helpers.GET;
import static play.test.Helpers.route;

/**
* This class ensure MatchController running well
* Referred :-
https://www.playframework.com/documentation/2.8.x/JavaFunctionalTest
*/
public class MatchControllerTest extends WithApplication {

   /**
```

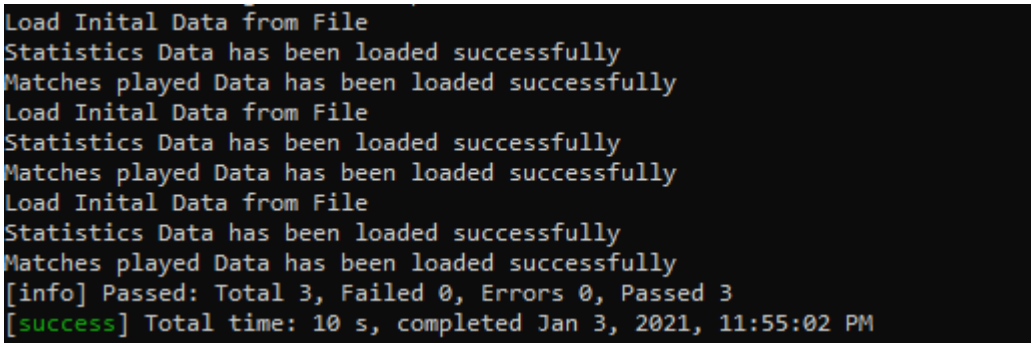
```
  * Override play app configuration for the test
  */
  @Override
  protected Application provideApplication() {
     return new GuiceApplicationBuilder()
 .configure("file.league_file_name", "LeagueFileTest.txt")
     .configure("file.match_file_name", "MatchFileTest.txt")
 .build();
  }

  /**
  * search match by date API Test , should be http 200 response code
  */
  @Test
  public void getMatchesByDateTest() {
     Http.RequestBuilder request = new Http.RequestBuilder()
           .method(GET)

.uri("http://localhost:9000/premierleague/match/matches?date=28/12/2020");

     Result result = route(app, request);
     assertEquals(OK, result.status());
  }

  /**
  * match list order by date ascending API Test , should be http 200 response code
  */
        @Test
  public void getMatchesOrderByDateTest() {
     Http.RequestBuilder request = new Http.RequestBuilder()
           .method(GET)
           .uri("http://localhost:9000/premierleague/match/matches/order-by-date");

     Result result = route(app, request);
     assertEquals(OK, result.status());
  }

  /**
  * random match API Test , should be http 200 response code
  */
        @Test
  public void getRandomMatchTest() {
     Http.RequestBuilder request = new Http.RequestBuilder()
           .method(GET)
           .uri("http://localhost:9000/premierleague/match/random");
```



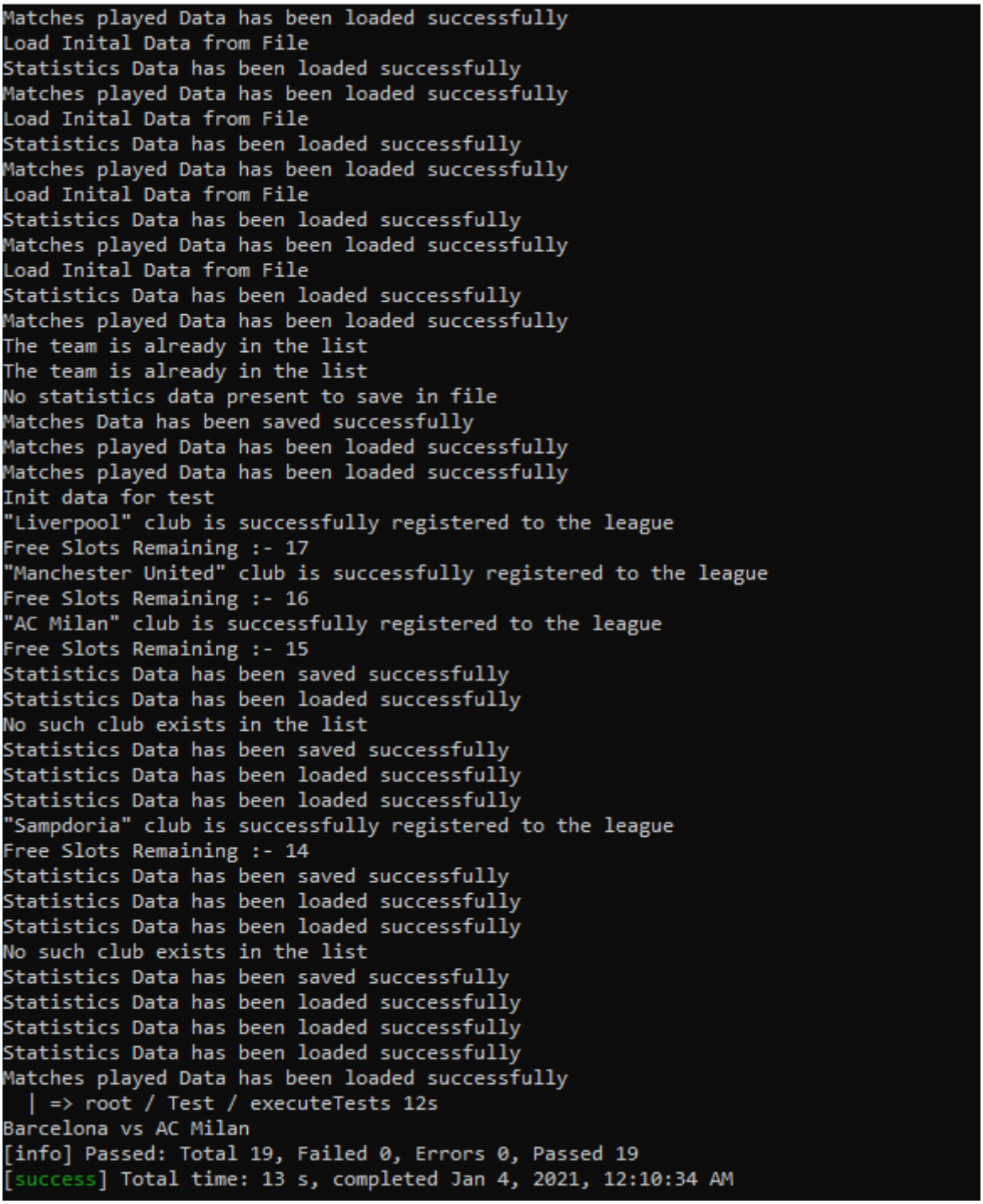
```
     Result result = route(app, request);
     assertEquals(OK, result.status());
```

```
    }
}
```

## Screenshot of MatchControllerTest class



Screenshot when all classes are run



# 4) UI Components - Angular

## 4.1)Main page (main-gui)

### 4.1.1) main-gui.component.html

- This consists the main page of GUI.

```
<div class="mainGui">
<h1>Welcome to Premier League Application Managament</h1>
</div>
```

### 4.1.2) main-gui.component.ts

```
import { Component, OnInit } from '@angular/core';

@Component({
 selector: 'app-main-gui',
 templateUrl: './main-gui.component.html',
 styleUrls: ['./main-gui.component.scss']
})
export class MainGuiComponent implements OnInit {
 constructor() { }

 ngOnInit() {
 }

}
```

## 4.2)Navigation Bar

### 4.2.1)app.component.html

```
<header class="header-fixed">
        <div class="header-limiter">
         <span class="toggle-btn" style="float:left; padding:6px; padding-right:16px;"
(click)="toogleSideBar()"  >
                <mat-icon *ngIf="!toogle; else close"
style="cursor:pointer">reorder</mat-icon>
         </span>
                 <a href="#"><img src="../assets/images/logo.png" alt="logo" /> </a>
        </div>
        <ng-template #close>
```

```
  <mat-icon style="cursor:pointer">close</mat-icon>
</ng-template>
</header>

<!-- to prevent the content of the page from jumping up -->
<div class="header-fixed-placeholder"></div>
<div id="sidebar" [class.active]="toogle">
  <div class="list">
    <!-- navigation bar options -->
    <div class="item"  [routerLink]="['/main-gui']" >- Home</div>
    <div class="item" (click)="toogleSideBar()" [routerLink]="['/first-gui']" >- VIEW
PREMIER LEAGUE TABLE</div>
    <div class="item" (click)="toogleSideBar()"[routerLink]="['/second-gui']">- PLAY A
RANDOM MATCH</div>
    <div class="item" (click)="toogleSideBar()" [routerLink]="['/third-gui']">- SEARCH
MATCHES PLAYED BY DATE</div>
    <div class="item" (click)="toogleSideBar()" [routerLink]="['/fourth-gui']">- VIEW AND
SORT MATCHES PLAYED BY DATE</div>
  </div>
</div>
<div class="place">
<router-outlet></router-outlet>
</div>
```

### 4.2.2) app.component.ts

```
import { Component } from '@angular/core';

@Component({
  selector: 'app-root',
  templateUrl: './app.component.html',
  styleUrls: ['./app.component.scss']
})
export class AppComponent {
  title = 'premier-league';
  toogle = true;

  //show or hide the navigation bar
  toogleSideBar():void{
            this.toogle = !this.toogle
  }
}
```

### 4.2.3) app-routing.module.ts

```
import { NgModule } from '@angular/core';
import { Routes, RouterModule } from '@angular/router';
import { ErrorComponent } from './error/error.component';
import { FirstGuiComponent } from './first-gui/first-gui.component';
import { FourthGuiComponent } from './fourth-gui/fourth-gui.component';
import { MainGuiComponent } from './main-gui/main-gui.component';
import { SecondGuiComponent } from './second-gui/second-gui.component';
import { ThirdGuiComponent } from './third-gui/third-gui.component';

//routes for Angular Premier League App
const routes: Routes = [
 {
  path: '',
  redirectTo: 'main-gui',
  pathMatch: 'full'
 },
 {
  path: 'main-gui',
  component: MainGuiComponent
 },
 {
  path: 'first-gui',
  component: FirstGuiComponent
 },
 {
  path: 'second-gui',
  component: SecondGuiComponent
 },
 {
  path: 'third-gui',
  component: ThirdGuiComponent
 },
 {
  path: 'fourth-gui',
  component: FourthGuiComponent
 },
 {
  path: '**',
  component: ErrorComponent
 }
];

@NgModule({
 imports: [RouterModule.forRoot(routes)],
 exports: [RouterModule]
})
```

```
export class AppRoutingModule { }
```

## 4.3) CSS and Interfaces

### 4.3.1) index.ts

```
//types for angular premier league app
export interface IScore {
   clubName: string;
   totalMatches: number;
   wins: number;
   draws: number;
   defeats: number;
   points: number;
   goalsScored: number;
   goalsReceived: number;
   goalsDifference: number;
}

export interface IClub {
   date: Date;
   home_club: string;
   home_score: number;
   away_score: number;
   away_club: string;
}

export interface IFootballClub {
   wins: number;
   draws: number;
   defeats: number;
   goalsScored: number;
   goalsReceived: number;
  goalsDifference: number;
   totalMatches: number;
   points: number;

   clubName: string;
   clubLocation: string;
   clubManager: string;
   clubFormedYear: number;
}

export interface IMatch {
   datePlayed: string;
   awayScore: number;
   awayTeam: IFootballClub;
```

```
    homeScore: number;
    homeTeam: IFootballClub;
}
```

### 4.3.2) index.html

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Premier League</title>
  <base href="/">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="icon" type="image/x-icon" href="assets/images/favicon.ico">
  <link
href="https://fonts.googleapis.com/css?family=Roboto:300,400,500&display=swap"
rel="stylesheet">
  <link rel="stylesheet"
        href="https://fonts.googleapis.com/css?family=Langar">
  <link href="https://fonts.googleapis.com/icon?family=Material+Icons"
rel="stylesheet">
  <style>
    body {
      background-image: url(assets/images/2017.jpg);
      background-repeat: round;
            color: white !important;
    }

    .table td {
      text-align: center;
            border-top: 1px solid #858b92 !important;
    }

          .table-striped tbody tr:nth-of-type(odd) {
    background-color: rgb(15 90 162) !important;
    color: white;
    font-weight: bold;
}

          .table-striped tbody tr:nth-of-type(even) {
    background-color: rgb(90 158 224 / 71%) !important;
    color: white;
    font-weight: bold;
}

          .table th {
          background: #104375;
          border-top: 4px solid #346492 !important;
          border-bottom: 4px solid #346492 !important;
```

```
        border-left: 4px solid #346492 !important;
        border-right: 4px solid #346492 !important;
        color: white;

        }

 .table thead th {
   text-align: center;
   font-size: 21px;
 }

 .container {
   text-align: center;
 }

 .score-card-list {
   display: flex;
   flex-direction: column;
   align-items: center;
 }

 .score-card {
   width: 550px;
   display: flex !important;
 }

 .home {
   flex: 1;
   text-align: center;
 }

 .away {
   flex: 1;
   text-align: center;
 }

 .score {
   width: 100px;
 }

        #sidebar {
position:fixed;
top: 12%;
left:-200px;
width:200px;
height:100%;
background:#104375;
transition:all 300ms linear;
```

```
  z-index: 3;
}
#sidebar.active {
  left:0px;
  z-index:3
}
.place{
    height: auto;
    transition: all 300ms ease-in-out;
}
#sidebar .toggle-btn {
  position:absolute;
  left:220px;
  top:10px;
  z-index: 3;
  cursor:pointer;
}
#sidebar .toggle-btn span {
  display:block;
  width:30px;
  height:5px;
  background:#151719;
  margin:5px 0px;
  cursor:pointer;
}
#sidebar div.list div.item {
  padding:15px 10px;
  color:#fcfcfc;
  text-transform:uppercase;
  font-size:15px;
  cursor:pointer;
  border: none;
  border-bottom: 1px solid #20568b;
}

#sidebar div.list div.item:hover {
 color: orange
}

#sidebar div.list div.item:focus {
 background-color: grey
}

.header-fixed {
        background-color:#104375;
        padding: 20px 40px;
        height: 80px;
        color: #ffffff;
```

```
	box-sizing: border-box;
	top:-100px;

	-webkit-transition:top 0.3s;
	transition:top 0.3s;
}

.header-fixed .header-limiter {
	max-width: 1200px;
	margin: 0 auto;
}

/*	The header placeholder. It is displayed when the header is fixed to the top of
the
	browser window, in order to prevent the content of the page from jumping up.
*/

.header-fixed-placeholder{
	height: 80px;
	display: none;
}

/* Logo */

.header-fixed .header-limiter h1 {
	float: left;
	font: normal 28px Cookie, Arial, Helvetica, sans-serif;
	line-height: 40px;
	margin: 0;
}

h2{
	line-height: 40px;
	font: normal 28px Langar;
	font-size: 3rem !important;
	text-align: center;
}

.header-fixed .header-limiter h1 span {
	color: #5383d3;
}

/* The navigation links */

.header-fixed .header-limiter a {
	color: #ffffff;
	text-decoration: none;
}
```

```
.header-fixed .header-limiter nav {
	font:16px Arial, Helvetica, sans-serif;
	line-height: 40px;
	float: right;
}

.header-fixed .header-limiter nav a{
	display: inline-block;
	padding: 0 5px;
	text-decoration:none;
	color: #ffffff;
	opacity: 0.9;
}

.header-fixed .header-limiter nav a:hover{
	opacity: 1;
}

.header-fixed .header-limiter nav a.selected {
	color: #608bd2;
	pointer-events: none;
	opacity: 1;
}

/* Fixed version of the header */

body.fixed .header-fixed {
	padding: 10px 40px;
	height: 50px;
	position: fixed;
	width: 100%;
	top: 0;
	left: 0;
	z-index: 1;
}

body.fixed .header-fixed-placeholder {
	display: block;
}

body.fixed .header-fixed .header-limiter h1 {
	font-size: 24px;
	line-height: 30px;
}
```



```
body.fixed .header-fixed .header-limiter nav {
	line-height: 28px;
```

```
        font-size: 13px;
}


/* Making the header responsive */

@media all and (max-width: 600px) {

        .header-fixed {
                padding: 20px 0;
                height: 75px;
        }

        .header-fixed .header-limiter h1 {
                float: none;
                margin: -8px 0 10px;
                text-align: center;
                font-size: 24px;
                line-height: 1;
        }

        .header-fixed .header-limiter nav {
                line-height: 1;
                float:none;
        }

        .header-fixed .header-limiter nav a {
                font-size: 13px;
        }

        body.fixed .header-fixed {
                display: none;
        }

}

.legend{
        border: 4px solid #346492;
   font-weight: bold;
   padding: 20px;
   text-align: initial;
   background-color: #104375;
}

.RedButton {
        box-shadow: 3px 4px 0px 0px #8a2a21;
        background-color:#c62d1f;
        border-radius:18px;
```

```
	border:1px solid #d02718;
	display:inline-block;
	cursor:pointer;
	color:#ffffff;
	font-family:Arial;
	font-size:17px;
	padding:7px 25px;
	text-decoration:none;
	text-shadow:0px 1px 0px #810e05;
}
.RedButton:hover {
	background-color:#f24437;
}
.RedButton:active {
	position:relative;
	top:1px;
}

.BlackButton {
	box-shadow: 3px 4px 0px 0px black;
	background-color:black;
	border-radius:18px;
	border:1px solid black;
	display:inline-block;
	cursor:pointer;
	color:#ffffff;
	font-family:Arial;
	font-size:17px;
	padding:7px 25px;
	text-decoration:none;
	text-shadow:0px 1px 0px #810e05;
}
.BlackButton:hover {
	background-color:gray;
}
.BlackButton:active {
	position:relative;
	top:1px;
}

.GreenButton {
	box-shadow: 3px 4px 0px 0px green;
	background-color:#20c997;
	border-radius:18px;
	border:1px solid green;
	display:inline-block;
	cursor:pointer;
	color:#ffffff;
```

```
	font-family:Arial;
	font-size:17px;
	padding:7px 25px;
	text-decoration:none;
	text-shadow:0px 1px 0px #810e05;
}
.GreenButton:hover {
	background-color:#7ead7e;
}
.GreenButton:active {
	position:relative;
	top:1px;
}

.BlueButton {
	box-shadow: 3px 4px 0px 0px #104375;
	background-color:#104375;
	border-radius:18px;
	border:1px solid #104375;
	display:inline-block;
	cursor:pointer;
	color:#ffffff;
	font-family:Arial;
	font-size:17px;
	padding:5px;
	text-decoration:none;
	text-shadow:0px 1px 0px #810e05;
}
.BlueButton:hover {
	background-color:#104375;
}
.BlueButton:active {
	position:relative;
	top:1px;
}

.mainGui{
}

h1 {
  -webkit-transition: all 3s ease-out;
       transition: all 3s ease-out;
  -webkit-animation: pulsate 8s ease-out infinite;
       animation: pulsate 8s ease-out infinite;
  position: absolute;
  top: 50%;
  left: 50%;
  -webkit-transform: translate(-50%, -50%);
```

```
        transform: translate(-50%, -50%);
  margin: 0;
  font-family: Helvetica, Arial, Sand-serif;
  font-weight: bold !important;
  font-size: 4em !important;
  color: white;
  text-align: center;
  text-transform: uppercase; }

h1:hover {
  color: white; }

/*** Keyframes - used with the "animation" property in the h1 ***/
@-webkit-keyframes pulsate {
  0%, 20% {
    -webkit-transform: translate(-50%, -50%) scale(1);
          transform: translate(-50%, -50%) scale(1);
    -webkit-filter: blur(5px);
          filter: blur(5px); }
  50%, 70% {
    -webkit-transform: translate(-50%, -50%) scale(1.5);
          transform: translate(-50%, -50%) scale(1.5);
    -webkit-filter: none;
          filter: none; }
  100% {
    -webkit-transform: translate(-50%, -50%) scale(1);
          transform: translate(-50%, -50%) scale(1);
    -webkit-filter: blur(5px);
          filter: blur(5px); } }
@keyframes pulsate {
  0%, 20% {
    -webkit-transform: translate(-50%, -50%) scale(1);
          transform: translate(-50%, -50%) scale(1);
    -webkit-filter: blur(5px);
          filter: blur(5px); }
  50%, 70% {
    -webkit-transform: translate(-50%, -50%) scale(1.5);
          transform: translate(-50%, -50%) scale(1.5);
    -webkit-filter: none;
          filter: none; }
  100% {
    -webkit-transform: translate(-50%, -50%) scale(1);
          transform: translate(-50%, -50%) scale(1);
    -webkit-filter: blur(5px);
          filter: blur(5px); } }
```



```
/*** Media queries for adjusting to different screen sizes ***/
@media only screen and (max-width: 600px) {
```

```
 h1 {
   font-size: 2em; }
}

#sidebar.active ~ .place {
   padding-left: 14%;
          transition:all 300ms linear;
}
#sidebar {
   padding:1%;
          transition:all 300ms linear;
}

.header-limiter img{
          float: left;
   width: auto;
   height: 71px;
   margin-top: -17px;
}

 </style>
</head>

<body>
 <app-root></app-root>
</body>

</html>
```

## 4.4) View Premier League Table (first-gui)

### 4.4.1) first-gui.component.html

1. Consists of Premier League statistics table which can be sorted according to Number of Wins,Number of Points and Number of Goals Scored.

```
<div class="container mt10">
 <div class="row">
   <div class="col-lg-9">
          <h2>Premier League Table</h2>
     <div class="table-responsive">
       <table class="table table-striped">
```

```
<!-- table headings -->
<thead>
        <tr>
         <th>#</th>
          <th>Club</th>
          <th>MP</th>
          <th>W</th>
          <th>D</th>
          <th>L</th>
          <th>GF</th>
          <th>GA</th>
          <th>GD</th>
          <th>Pts</th>

        </tr>
       </thead>
       <tbody>
         <!-- table data -->
        <tr *ngFor="let score of scores; let i = index">
          <td>{{i + 1}}</td>
          <td>{{ score.clubName }}</td>
          <td> {{ score.totalMatches }} </td>
          <td> {{ score.wins }} </td>
          <td> {{ score.draws }} </td>
          <td> {{ score.defeats }} </td>
          <td> {{ score.goalsScored }} </td>
          <td> {{ score.goalsReceived }} </td>
          <td> {{ score.goalsDifference }} </td>
           <td> {{ score.points }} </td>
        </tr>
       </tbody>
     </table>
   </div>
         <div class="row mt10">
                <div class="col-lg-12">
                 <div class="float-right">
                        <button class="RedButton mx10" (click)="sort('wins')">Sort By
Wins</button>
                        <button class="BlackButton mx10" (click)="sort('points')">Sort By
Points</button>
                        <button class="GreenButton mx10" (click)="sort('goals')">Sort By
Goals Scored</button>
                  </div>
                </div>
          </div>
  </div>
        <!-- legend displayed on right -->
        <div class="col-lg-3">
```

```
                <div class="legend">
                        <p>MP - Matches Played</p>
                        <p>W - Wins</p>
                        <p>D - Draws</p>
                        <p>L - Loss (Defeat)</p>
                        <p>GF - Goals Scored</p>
                        <p>GA - Goals Received</p>
                        <p>GD - Goals Difference</p>
                        <p>Pts - Points</p>

                </div>
        </div>
 </div>

</div>
```

### 4.4.2) first-gui.component.ts

```
import { Component, OnInit } from '@angular/core';
import { IMatch } from '../models';
import { DataService } from '../services/data.service';

@Component({
 selector: 'app-fourth-gui',
 templateUrl: './fourth-gui.component.html',
 styleUrls: ['./fourth-gui.component.scss']
})
export class FourthGuiComponent implements OnInit {

 matches : IMatch[] = [];

 constructor(private dataService: DataService) {
 }

 // Sending request to get teams
 ngOnInit() {
  this.dataService.sendGetRequest('match/matches').subscribe((data: IMatch[]) => {
   this.matches = data;
  });
 }

 // Sending request to get teams by order to display data in table
 sort() {
  this.dataService.sendGetRequest('match/matches/order-by-date').subscribe((data:
IMatch[]) => {
   this.matches = data;
  });
 }
}
```

## 4.5) Play A Random Match (second-gui)

### 4.5.1) second-gui.component.html

```
<div class="container mt10">
   <div class="row">
      <div class="col-lg-9">
                  <h2>Matches Played</h2>
         <div class="float-right py10">
            <button class="GreenButton" (click)="randomMatch()">Play A Random
Match</button>
         </div>
         <div class="table-responsive">
            <table class="table table-striped">
               <!-- table headings -->
               <thead>
                  <tr>
                     <th>#</th>
                     <th>Club</th>
                     <th>MP</th>
                     <th>W</th>
                     <th>D</th>
                     <th>L</th>
                     <th>GF</th>
                     <th>GA</th>
                     <th>GD</th>
                     <th>Pts</th>
                  </tr>
               </thead>
               <tbody>
                  <!-- table data -->
                  <tr *ngFor="let score of scores; let i = index">
                     <td>{{i + 1}}</td>
                     <td>{{ score.clubName }}</td>
                     <td> {{ score.totalMatches }} </td>
                     <td> {{ score.wins }} </td>
                     <td> {{ score.draws }} </td>
                     <td> {{ score.defeats }} </td>
                     <td> {{ score.goalsScored }} </td>
                     <td> {{ score.goalsReceived }} </td>
                     <td> {{ score.goalsDifference }} </td>
                     <td> {{ score.points }} </td>
                  </tr>
               </tbody>

            </table>
```

```
        </div>
      </div>
                <!-- legend displayed on right -->
                <div class="col-lg-3">
                <div class="legend">
                        <p>MP - Matches Played</p>
                        <p>W - Wins</p>
                        <p>D - Draws</p>
                        <p>L - Loss (Defeat)</p>
                        <p>GF - Goals Scored</p>
                        <p>GA - Goals Received</p>
                        <p>GD - Goals Difference</p>
<p>Pts - Points</p>
                </div>
          </div>
    </div>
</div>
```

### 4.5.2) second-gui.component.ts

```
import { Component, OnInit } from '@angular/core';
import { MatDialog } from '@angular/material';
import { IMatch, IScore } from '../models';
import { SecondGuiModalComponent } from '../second-gui-modal/second-gui-
modal.component';
import { DataService } from '../services/data.service';
@Component({
  selector: 'app-second-gui',
  templateUrl: './second-gui.component.html',
  styleUrls: ['./second-gui.component.scss']
})
export class SecondGuiComponent implements OnInit {

  scores: IScore[] = [];

  constructor(
    public dialog: MatDialog, private dataService: DataService
  ) { }

  ngOnInit() {
    this.getTeams();
  }

  // Sending request to get teams order by points for display in data table
  private getTeams(): void {
    this.dataService.sendGetRequest('team/teams', { orderBy: 'points' }).subscribe((data:
IScore[]) => {
      this.scores = data;
```

```
  });
 }

 // Sending request to get random method to display in Mat Dialog
 randomMatch() {
  this.dataService.sendGetRequest('match/random').subscribe((data: IMatch) => {
   this.dialog.open(SecondGuiModalComponent, {
    data
   }).afterClosed().subscribe((evt: any) => {
    this.getTeams();
   });
  });
 }
}
```

### 4.5.3) second-gui-modal.component.html

```
<div class="container">
                <button mat-icon-button mat-dialog-close aria-label="close"
style="float:right; margin-top:-21px;">
                        <mat-icon>close</mat-icon>
                </button>
                 <h5 style="text-align: initial">Date: {{data.datePlayed}}</h5>
    <div class="score-card">
      <h3 class="home">{{data.homeTeam}}</h3>
      <h3 class="score"></h3>
      <h3 class="away">{{data.awayTeam}}</h3>
    </div>
                <div class="score-card">
                   <h3 class="home"></h3>
      <h3 class="score">vs</h3>
      <h3 class="away"></h3>
    </div>

    <div class="score-card">
      <h3 class="home">{{data.homeScore}}</h3>
      <h3 class="score"></h3>
      <h3 class="away">{{data.awayScore}}</h3>
    </div>

</div>
```

### 4.5.4) second-gui-modal.component.ts

```
import { Component, Inject, OnInit } from '@angular/core';
import { MatDialogRef, MAT_DIALOG_DATA } from '@angular/material';
```

```
@Component({
 selector: 'app-second-gui-modal',
 templateUrl: './second-gui-modal.component.html',
 styleUrls: ['./second-gui-modal.component.scss']
})
export class SecondGuiModalComponent implements OnInit {

 // Configuration for Mat Dialog Component
 constructor(
  public dialogRef: MatDialogRef<SecondGuiModalComponent>,
  @Inject(MAT_DIALOG_DATA) public data: SecondGuiModalComponent
 ) { }

 ngOnInit() {
 }

}
```

## 4.6) SEARCH MATCHES PLAYED WITH DATE (third-gui)

### 4.6.1) third-gui.component.html

```
<div class="container mt10">
   <div class="row">
      <div class="col-lg-12">
                              <h2>Matches Played</h2>
         <div class="row py10">
            <!-- text box for day -->
            <div class="col-lg-2">
               <input placeholder="Day" type="number" min="1" max="31" class="form-
control" name="day" [(ngModel)]="day">
            </div>
            <!-- text box for month -->
            <div class="col-lg-2">
               <input placeholder="Month" type="number" min="1" max="12"
class="form-control" name="month" [(ngModel)]="month">
            </div>
            <!-- text box for year -->
            <div class="col-lg-2">
               <input placeholder="Year" type="number" min="1999" max="2030"
class="form-control" name="year" [(ngModel)]="year">
            </div>
            <div class="col-lg-2">
               <div class="">
                  <button class="GreenButton" (click)="searchByDate()">Search</button>
               </div>
```

```
            </div>
                                    <div class="col-lg-2">
              <div class="">
                                                    <button class="RedButton"
(click)="clear()">Clear Date</button>
              </div>
            </div>
          </div>

          <div class="table-responsive">
            <table class="table table-striped">
              <!-- table headings -->
              <thead>
                <tr>
                  <th>Date</th>
                  <th>Home Club</th>
                  <th>Home Score</th>
                  <th>Away Score</th>
                  <th>Away Club</th>
                </tr>
              </thead>
              <tbody>
                <!-- table data -->
                <tr *ngFor="let match of matches">
                  <td> {{ match.datePlayed}} </td>
                  <td> {{ match.homeTeam }} </td>
                  <td> {{ match.homeScore }} </td>
                  <td> {{ match.awayScore }} </td>
                  <td> {{ match.awayTeam }} </td>
                </tr>
              </tbody>

            </table>
          </div>

        </div>
      </div>
    </div>
```

### 4.6.2) third-gui.component.ts

```
import { Component, OnInit } from '@angular/core';
import { IMatch } from '../models';
import { DataService } from '../services/data.service';
import { DatePipe } from "@angular/common";

@Component({
```

```
 selector: 'app-third-gui',
 templateUrl: './third-gui.component.html',
 styleUrls: ['./third-gui.component.scss']
})
export class ThirdGuiComponent implements OnInit {

 matches: IMatch[] = [];
 day: number;
 month: number;
 year: number;

 constructor(private dataService: DataService) { }

 ngOnInit() {
  this.matchRequest();
 }

 //if the date is empty get all matches and if it is not empty try to get the matches by
date parameter
 searchByDate() {
        let date;
        if(!(this.day || this.month || this.year)){
                date = "";
        }
  else if(this.validateDate()){
                date = `${this.day}/${this.month}/${this.year}`;
        }
  this.matchRequest(date);
 }

 // Sending request to get matches by date
 private matchRequest(date: string = ""): void {
  this.dataService.sendGetRequest('match/matches', { date })
    .subscribe((data: IMatch[]) => {
     this.matches = data;
    }, (error) => {
     if (error.status === 409)
       alert(error.error)
    });
 }

 // Validation of fields day,month and year before sending the match request
 private validateDate(): boolean{
         let dateString = `${this.day}/${this.month}/${this.year}`;

         if(dateString.includes("undefined") || dateString.includes("null") ){
                 alert("Fill the rest of fields")
                 return false;
```

```
            }
            if(!(this.day >= 1 && this.day <= 31)){
                    alert("Days are between 1 and 31")
                    return false;
            }

            if(!(this.month >= 1 && this.month <= 12)){
                    alert("Months are between 1 and 12")
                    return false;
            }

            if(!(this.year >= 1993 && this.year <= 2040)){
                    alert("Years are between 1993 and 2040")
                    return false;
            }
            return true;
  }
  // clear all the text fields
  private clear():void{
            this.day = null;
            this.month = null;
            this.year = null;
  }
}
```

## 4.7) VIEW AND SORT MATCHES PLAYED WITH DATE (fourth-gui)

### 4.7.1) fourth-gui.component.html

```
<div class="container mt10">
   <div class="row">
      <div class="col-lg-12">
                  <h2>Matches Played</h2>
         <div class="py10 float-right">
            <button class="GreenButton" (click)="sort()">Sort in Ascending
order</button>
         </div>
         <div class="table-responsive">
            <table class="table table-striped">
               <!-- table headings -->
               <thead>
                  <tr>
                     <th>Date</th>
                     <th>Home Club</th>
                     <th>Home Score</th>
                     <th>Away Score</th>
```

```
                <th>Away Club</th>
              </tr>
            </thead>
            <tbody>
              <!-- table data -->
              <tr *ngFor="let match of matches">
                <td> {{ match.datePlayed}} </td>
                <td> {{ match.homeTeam }} </td>
                <td> {{ match.homeScore }} </td>
                <td> {{ match.awayScore }} </td>
                <td> {{ match.awayTeam }} </td>
              </tr>
            </tbody>

          </table>
        </div>

      </div>
    </div>
  </div>
```

### 4.7.2) fourth-gui.component.ts

```
import { Component, OnInit } from '@angular/core';
import { IMatch } from '../models';
import { DataService } from '../services/data.service';

@Component({
 selector: 'app-fourth-gui',
 templateUrl: './fourth-gui.component.html',
 styleUrls: ['./fourth-gui.component.scss']
})
export class FourthGuiComponent implements OnInit {

 matches : IMatch[] = [];

 constructor(private dataService: DataService) {
 }

 //Sending request to get all matches to display in data table
 ngOnInit() {
  this.dataService.sendGetRequest('match/matches').subscribe((data: IMatch[]) => {
   this.matches = data;
  });
 }
```

```
//Sending request to get all matches in ascending order of date
sort() {
   this.dataService.sendGetRequest('match/matches/order-by-date').subscribe((data:
IMatch[]) => {
     this.matches = data;
   });
  }
}
```

## 4.8) data.service.ts

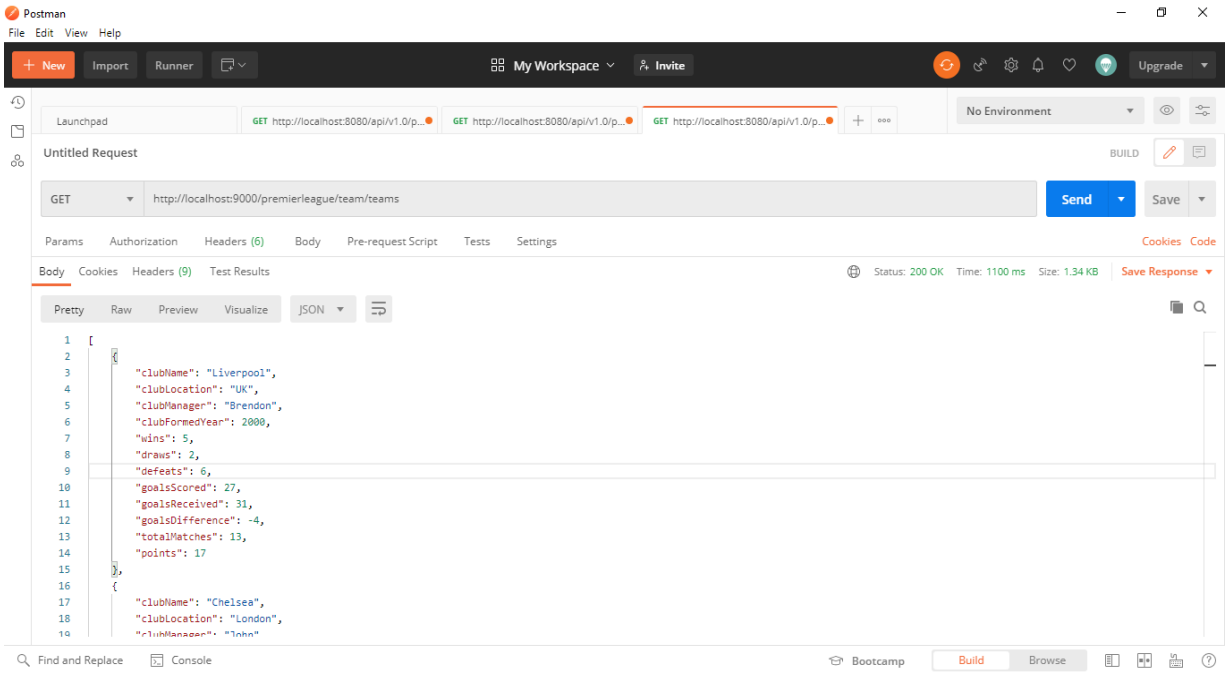
```
import { Injectable } from '@angular/core';
import { HttpClient } from '@angular/common/http';

@Injectable({
 providedIn: 'root'
})
export class DataService {

 private REST_API_SERVER = 'http://localhost:9000/premierleague/';

 constructor(private httpClient: HttpClient) {
 }
 //wrapping the Angular HTTP client get request
 public sendGetRequest(url: string, params?: any) {
   return this.httpClient.get(`${this.REST_API_SERVER}${url}`, {
     params: params
   });
 }
}
```

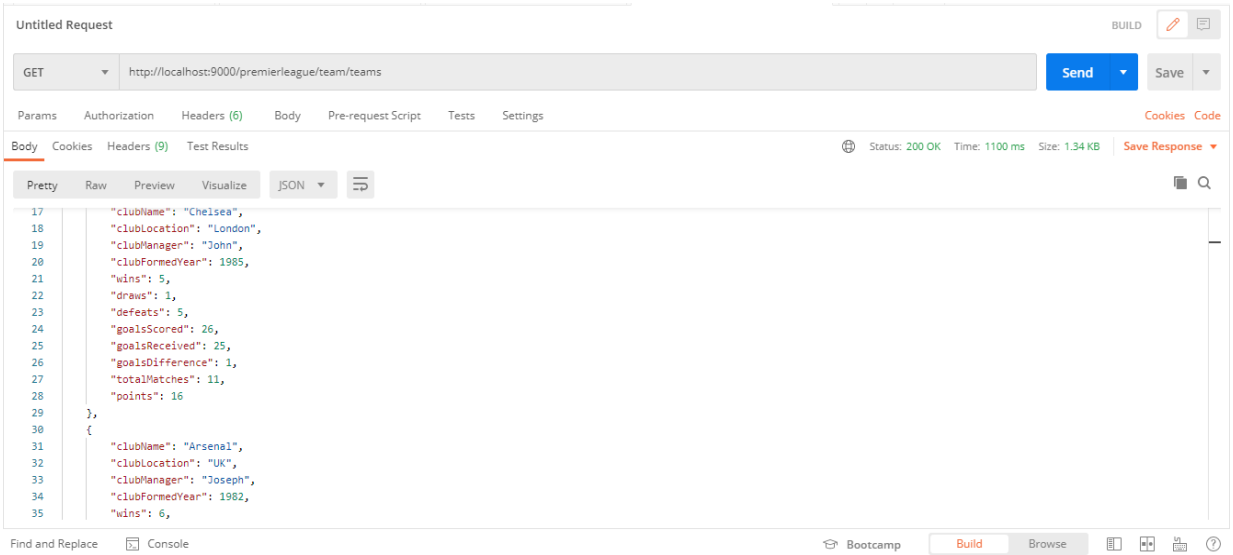
# 5) Routes – Play Framework

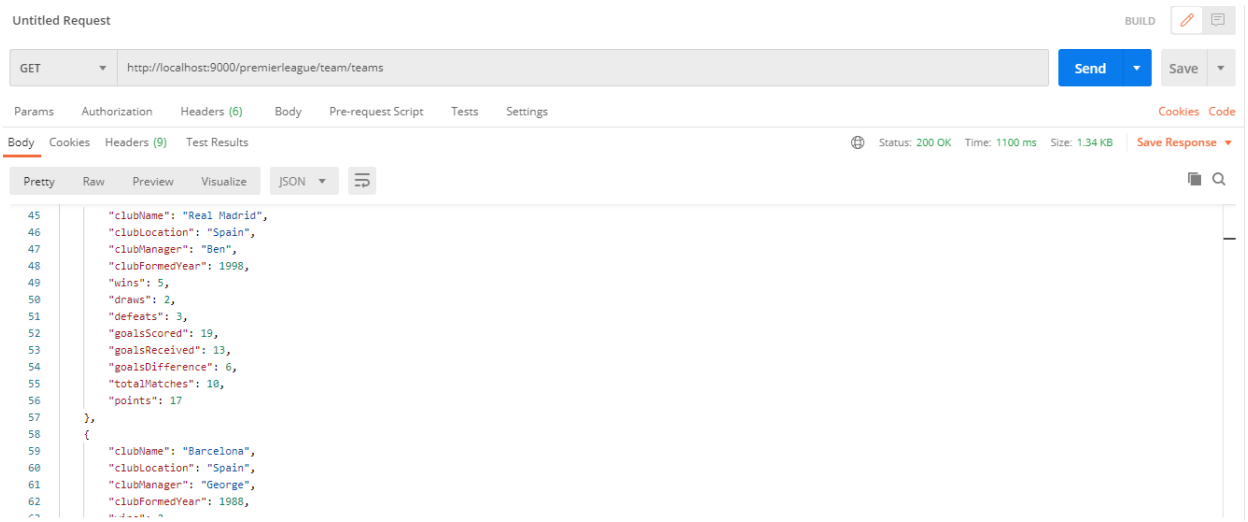
## 5.1)Screenshots of using Postman to test the REST APIs

1. # TEAMS

GET /premierleague/team/teams
controllers.team.TeamController.getTeams(orderBy:String ?= "")

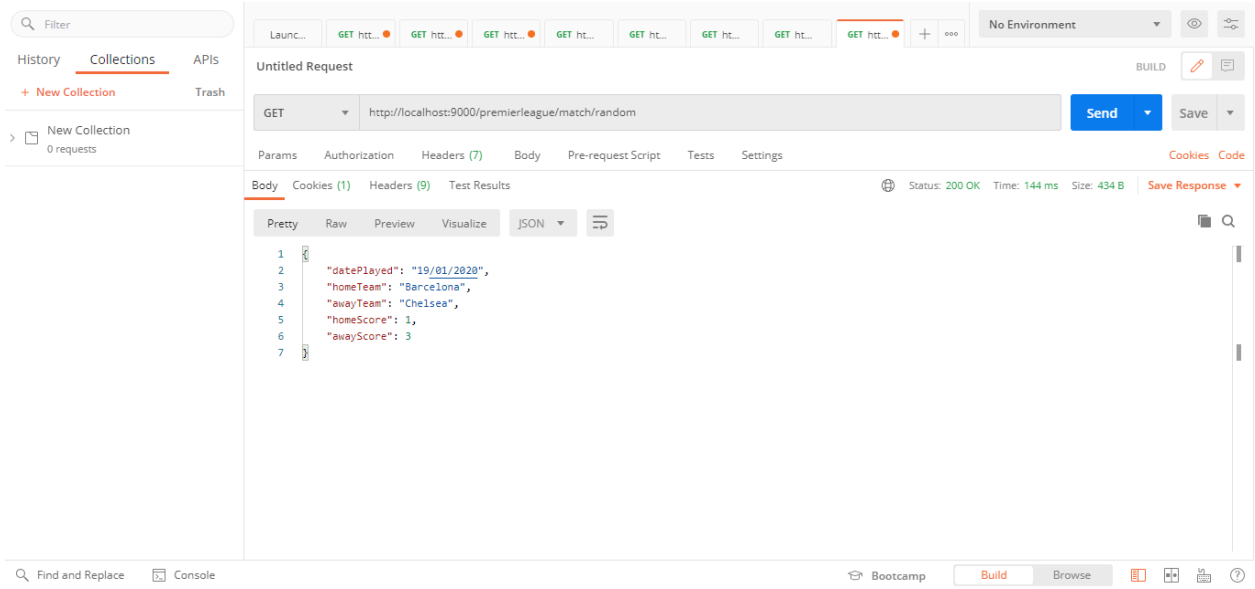





2. # MATCH

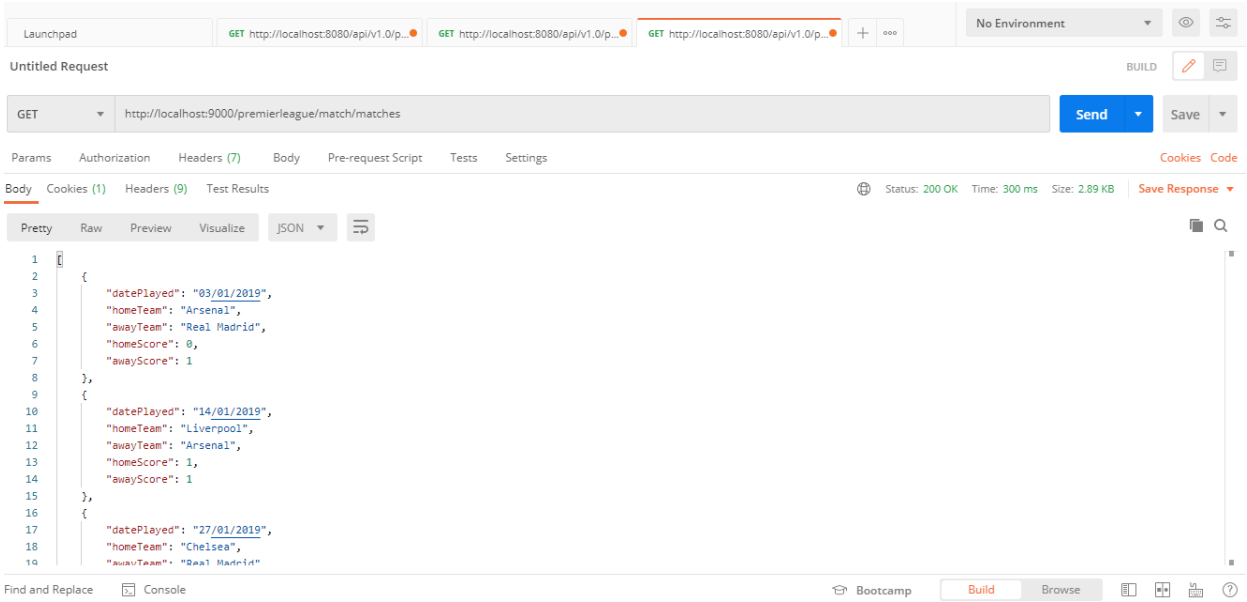
GET /premierleague/match/random
controllers.matches.MatchController.generateRandomMatch()



3.

GET /premierleague/match/matches

controllers.matches.MatchController.findMatchesByDate(date: String ?= "")




4.

GET /premierleague/match/matches/order-by-date

controllers.matches.MatchController.getMatchesOrderAscByDate()